

Detecting Voltage Sinusoidal Harmonics using Artificial Intelligence/ Machine Learning Techniques



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Supervisor: Dr. Muhammad Yousif

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I dedicate this thesis titled “Detecting Voltage Harmonics using Artificial Intelligence Techniques” to the professors, supervisor, senior colleagues, and administration of the U.S.-Pakistan Center for Advanced Studies in Energy, whose support and guidance have been contributory in completing this Master's research manuscript.

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LIST OF SYMBOL, ABBREVIATIONS AND ACRONYMS

RES	Renewable Energy Sources
PQ	Power Quality
ESP	Electric Submersible Pump
VFD	Variable Frequency Drives
AC	Alternating Current
PCC	Point of Common Coupling
AI	Artificial Intelligence
ML	Machine Learning
DG	Distributed Generation
DC	Direct Current
PEC	Power Electronic Components
EV	Electrical Vehicles
SVM	Support Vector Machines
KNN	K-Nearest Neighbors
NN	Neural Network
DT	Decision Tree
LR	Logistic Regression
RM	Random Forest
GB	Gradient Boosting
ANN	Artificial Neural Network
IEEE	Institute of Electrical and Electronics Engineers
IEC	International Electrotechnical Commission
STATCOM	Static Synchronous Compensator
UPQC	Unified Power Quality Conditioner

FFT	Fast Fourier Transform
THD	Total Harmonic Distortion

ABSTRACT

In advanced control frameworks, integrating Renewable Energy Sources (RES) has posed significant challenges. These sustainable energy sources, with their reduced greenhouse gas emissions, face obstacles in determining a dominant role due to various limitations. The presence of non-linear loads within electrical systems adds complexity, leading to constrained conditions. When these electrical systems interconnect with smart grids, these challenges become even more complex and present crucial impediments. Power quality (PQ) within electrical systems has become a predominant issue, causing financial losses due to PQ degradations. Therefore, approaches involving harmonic frequency waves need to be considered in these Alternating Current (AC) sinusoidal frequency waves. Furthermore, PQ disturbances need to be identified with precision and accuracy when performing Machine Learning (ML) algorithms efficiently. In contemporary power systems, the significant PQ issue of noise and waveform distortion may arise from the integration of RES, power electronic devices, and nonlinear loads, resulting in poor performance. Consequently, they are predominant due to their negative impact on power quality. As a result, there is growing interest in developing the application of Artificial Intelligence (AI) approaches in the predictive model. The advantage of AI approaches lies in their ability to analyze harmonics in electrical control systems. Harmonic source classification and data clustering are also facilitated by these methods. MATLAB Simulink and the Orange Data Mining Program have implemented detection and analysis of harmonic source classification and data clustering. Furthermore, AI techniques are applied to analyze harmonics in control systems as well. Thus, it may be an idea to identify these harmonic cases.

Keywords: Power Quality, Sinusoidal Harmonics, Artificial Intelligence Techniques

CHAPTER 1: INTRODUCTION

The application of nonlinearity in power circuits, whether due to the power supply or the load, is responsible for generating harmonic and waveform distortions. With power systems undergoing a transition from centralized to decentralized Renewable Energy Sources (RES) generation, there is a potential enhancement in power system dependability. Consequently, Power Quality (PQ) and controllability methods become crucial [1].

Power quality encompasses the measurement, analysis, and enhancement of a sinusoidal waveform at its rated voltage and frequency, significantly impacting a power system's performance and cost. PQ disturbances, such as overvoltage occurrences exceeding design limits, can result in damage to machinery or necessitate Power Factor (PF) correction. Additionally, transient voltage spikes caused by data processing errors or losses pose risks to equipment [2].

Overheating in equipment and voltage swells can lead to control issues and hardware breakdowns, as these spikes and transients are having high-energy bursts occurring within milliseconds. The advantageous effects of Distributed Generations (DGs) depend heavily on their deployment and operation. The introduction of DGs may introduce additional PQ, control difficulties, and stance challenges to electrical networks. Inverters play a crucial role in managing the electricity injected into the grid by converting Direct Current (DC) voltages to synchronized AC voltages [1].

Voltage stressors in power systems are classified based on their impact on protective devices or insulation, as per standards like IEC 60071 and IEEE Standard 1894. These stressors manifest in various classes observed at various stages of electricity flow, particularly in distribution and transmission phases [3].

One cause of voltage stressors includes temporary front overvoltage, characterized by short-term voltage spikes resulting from switching operations or transient events. Another cause is fast front overvoltage, induced by lightning strikes or other external factors. Similarly, very fast front overvoltage represents a rapid escalation in voltage levels, often associated with intense lightning activity or severe electrical disturbances [3].

The power system's supply restoration process, particularly concerning power quality, involves managing current transitions between operational points known as the Normal Open Point and the Soft Open Point. This dynamic process can lead to overcurrent conditions within the electricity transmission sector, posing risks to critical components such as power semiconductors [4].

Overcurrent events pose significant risks to power semiconductors, vital for various power system operations, including voltage control and power conversion. Excessive current flow through semiconductor devices can lead to performance degradation or permanent damage, resulting in thermal stress [4].

The widespread use of power electronic converter-fed devices such as computers, adjustable speed drives (ASDs), and light-emitting diodes (LEDs) in industrial applications and distribution systems has elevated power quality as a primary concern for energy firms and network operators. However, the nonlinear behavior of these devices in control systems presents challenges, with current transients revealing numerous flaws in the control system's voltage and current regulation [5].

In recent years, electric utilities have focused more on concerns about power quality disturbances, particularly those causing low voltage swell and sag [6]. Network disturbances, such as voltage droops from short circuit failures or frequency variations due to generation and consumption level disparities, complicate the analysis of grid-connected renewable energy sources. Inverters play a pivotal role in such scenarios, providing reactive control to increase voltage magnitude during voltage sags and eliminating sudden voltage spikes. Strict measures are in place to prevent overvoltages and undervoltages [7].

Furthermore, many electrical failures in Electric Submersible Pump (ESP) systems, for example, stem from poor Variable Frequency Drive (VFD) equipment output power quality. These failures, often exacerbated by high-voltage stress on cables, result in efficiency losses for operators. Common-mode and differential noises, along with other unintended consequences from subpar VFD output, pose immediate risks to device integrity. Proper measures must be taken to limit voltage and current noises produced by VFD topologies[8].

Power networks encounter significant challenges due to declines in power quality, leading to improper operation, decreased device efficiency, and shortened component lifespans. The presence of inductors and capacitors induces resonance in certain harmonics, intensifying voltage distortion in distribution networks. Microgrids face challenges like power fluctuation control and sustaining power quality in both grid-connected and island modes, contributing to power quality distortion [9].

The increasing prevalence of current harmonics resulting from the rising use of nonlinear electrical equipment leads to power loss and affects power quality. Inter-harmonics for instance, even at low amplitude levels, can cause voltage fluctuations, flickering lights, and sub-synchronous oscillations [10]. These harmonics should be vital in terms of the voltage and frequency too.

Therefore, understanding power quality through the gathering, analysis, and interpretation of data is imperative. Accurate detection and classification of disturbances enable proactive control actions [2].

Harmonics in alternating current (AC) sinusoidal waves involve the repetition of signal patterns across subsequent phases. These patterns, occurring in both voltage and current units, are represented by multiple integers (n) corresponding to the frequency of these electrical quantities [11].

The analysis of harmonics is a crucial endeavor in power systems engineering, involving the calculation of magnitudes and phases of periodic waveforms. Fourier's theorem provides a mathematical framework for understanding the complex interplay of harmonic components within electrical systems by decomposing periodic waveforms into the sum of sine waves [11].

1.1. Harmonics Definition and Fourier Series

When power electronics converters are active in manufacturing centers such as steel plants for an example, the electrical output often manifests as high-order harmonic currents and distorted sinusoidal waves [12]. The term "harmonics" in physics and engineering refers to the repetition of the original scientific content in subsequent phases, often involving new features and analyses. The fundamental and harmonic frequencies are represented by multiple numbers, as illustrated in Figure 1. The frequency of these voltage and current units, conveyed through AC sine waves, introduces the notation of harmonics.

To illustrate the concept of harmonic behavior, consider a string vibrating at different frequencies.

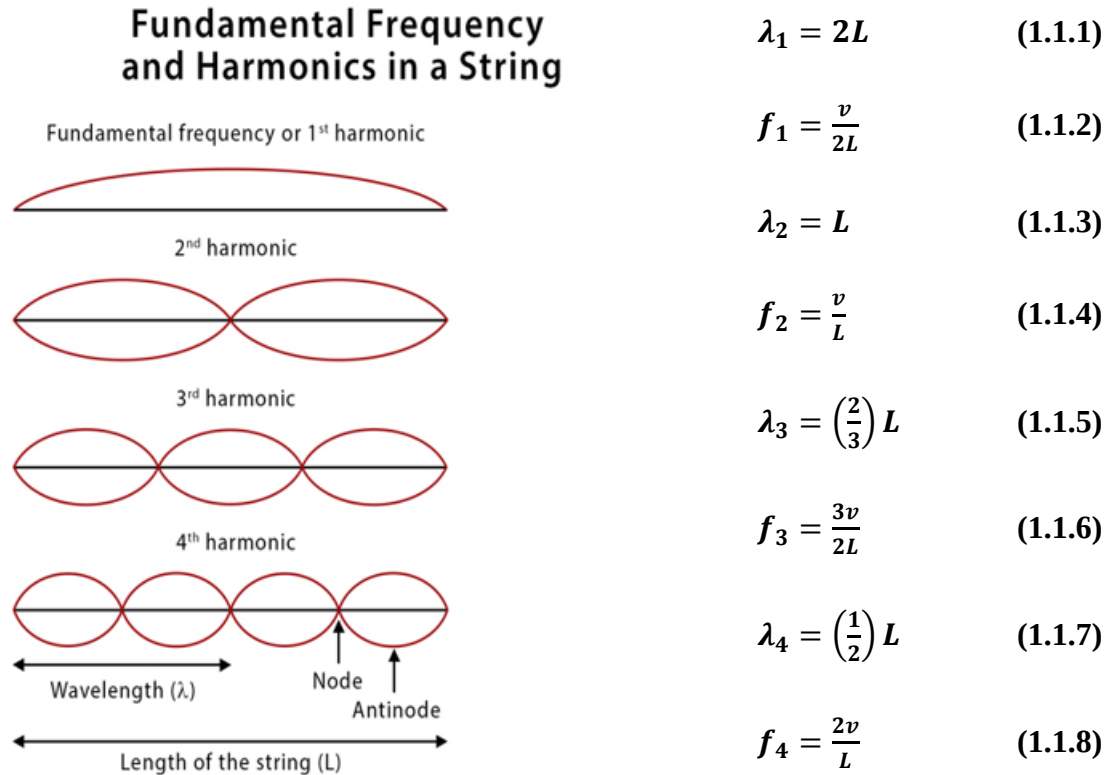


Figure 1: Harmonics in Demonstration

Fundamental Frequency (1st Harmonic): This is the least shape of string vibration where the whole length of the string vibrates as a single portion, coming about in a wavelength (λ_1) twice the length of the string ($2L$). The frequency (f_1) is decided by the speed of the wave (v) isolated by twice the length of the string ($2L$).

2nd Harmonic: The string vibrates in two fragments, making a design with one extra node in the center. The wavelength (λ_2) is rise to the length of the string (L), and the frequency (f_2) is the speed of the wave (v) separated by the length of the string (L).

3rd Harmonic: The string vibrates in three sections, with two extra nodes. The wavelength (λ_3) is two-thirds of the string's length ($2/3L$), and the frequency (f_3) is three times the speed of the wave (v) partitioned by twice the length of the string ($2L$).

4th Harmonic: The string vibrates in four segments, with three additional nodes. The wavelength (λ_4) is half the length of the string ($1/2L$), and the frequency (f_4) is four times the speed of the wave (v) isolated by the length of the string (L).

In each harmonic, nodes are points where the string does not move, and antinodes are points where the string moves the most.

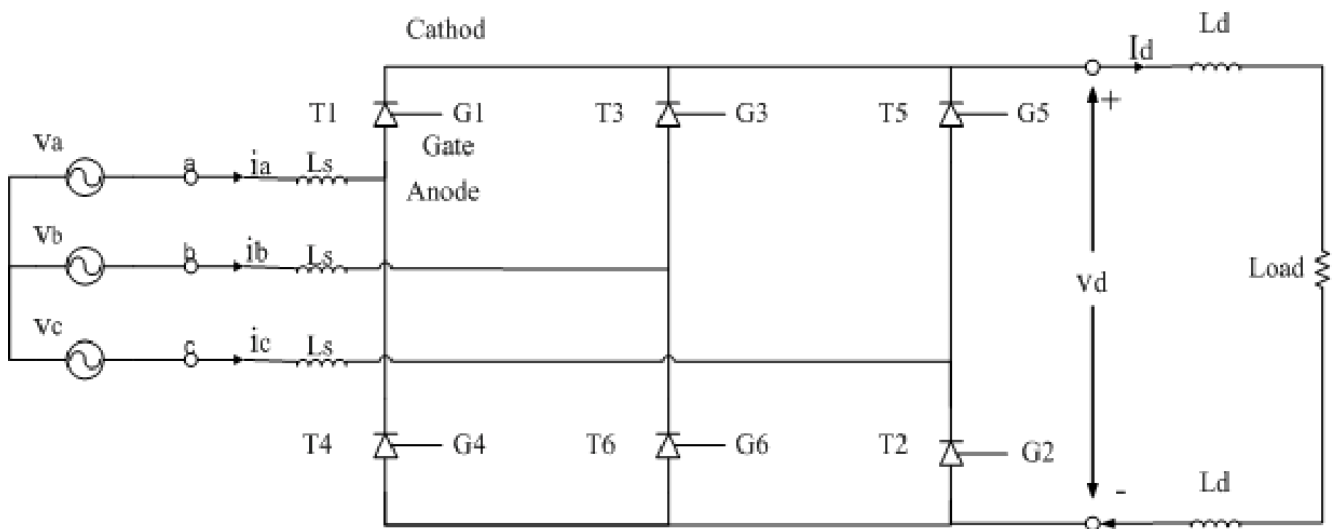


Figure 2: Six-Pulse Thyristor Rectifier

As shown in Figure 2, a six-pulse rectifier is a type of electrical converter that changes AC to DC using six diodes or thyristors. Harmonic analysis of a six-pulse rectifier involves studying the non-sinusoidal components generated due to the rectification process. These harmonics, multiples of the fundamental frequency, can cause distortion in the power system, with the most significant harmonics being the 5th and 7th, and their multiples [13].

- ❖ Diodes/Thyristors (T1-T6): Represented by the standard diode symbol, with anode and cathode terminals.
- ❖ Gate Signals (G1-G6): Indicate the control signals for the thyristors.
- ❖ Input Voltages (Va, Vb, Vc): The three-phase AC input voltages.
- ❖ Currents (ia, ib, ic): The currents flowing through each phase.
- ❖ DC Output Voltage (Vd): The resulting DC voltage across the load (Ld).

Six-pulse rectifiers find usage in medium to high-power applications such as variable speed drives, industrial power supplies, and DC motor controls, offering a relatively simple design and reduced harmonic distortion compared to single-phase rectifiers [14].

This phenomenon resembles the mathematical theory known as the Fourier Series. These equations (equation 1.1.9, equation 1.1.10, equation 1.1.11, and equation 1.1.12) correlate with AC wavelength curves, resulting in the generation of multiple harmonics. The Fourier Series equation encompasses three factors, as depicted in these equations. The term 'a₀' (Eq 1.1.10) denotes the average wavelength, while 'a_n' (Eq 1.1.11) and 'b_n' (Eq 1.1.12) represent the even and odd components respectively of the sinusoidal wave method. These factors contribute to the harmonic patterns observed in the AC sinusoidal curve as the three parameters regarding for AC sinusoidal harmonics.

$$\text{Fourier Series} = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega_1 t) + b_n \sin(n\omega_1 t)) \quad (1.1.9)$$

$$a_0 = \frac{1}{T} \int_0^T v(t) dt \quad (1.1.10)$$

$$a_n = \frac{2}{T} \int_0^T v(t) \cos(n\omega_1 t) dt \quad (1.1.11)$$

$$b_n = \frac{2}{T} \int_0^T v(t) \sin(n\omega_1 t) dt \quad (1.1.12)$$

The Fourier series represents a periodic function as a voltage (v) sum of sine and cosine functions which are harmonics of the fundamental frequency. The 'a₀' is the coefficient of the average value of the function over one period. This coefficient is calculated as the integral of the function over the period divided by the period length. The 'a_n' coefficients correspond to the cosine terms in the series, measuring the amplitude of even multiples of the fundamental frequency. The integral formula with period time (T) involves multiplying the function by (cos(n) × ω₁) – n: number of wave periods, ω: angular frequency and integrating over the period. Similarly, 'b_n' coefficients represent the odd harmonics, measured with the function multiplied by (sin(n) × ω₁).

1.2. Advantages and Failures in AC Sinusoidal Curves

Some advantages of AC sinusoidal harmonic waves lie in their application in signal processing and wavelet transformation techniques. These methods are valuable for illustrating signal behavior, particularly in applications such as radar signaling and frequency tweaking in communication systems. Additionally, they play a crucial role in adjusting amplitude or frequency patterns for radio stations. Furthermore, variations in frequency and amplitude units within AC sinusoidal curves can contribute to improved efficiency and effectiveness in electrical equipment and control electronics.

Despite these positive applications, the occurrence of harmonic frequencies presents several challenges. One major downside is their impact on power quality components, leading to poor or inadequate PQ behavior in the electrical system. Consequently, this is a significant drawback associated with harmonic frequency distortion in AC sinusoidal waveforms. Another adverse effect is the increased heating of cables and electrical transformers, which can reduce the efficiency and lifespan of electrical components. Resonance in the electrical system generates multiple harmonics, affecting system functions directly and potentially causing damage.

Excessive harmonics can result in various undesirable effects, including increased power losses, equipment overheating, voltage and current waveform distortions, interference with communication systems, and reduced power quality in electrical systems. In such cases, harmonics are considered negative or undesirable, and efforts are made to mitigate their effects.

Harmonics can adversely affect the performance of control devices, converters, and other equipment, leading to increased heating, reduced efficiency, and additional stresses on components, potentially shortening their lifespan.

1.3. Power Quality Risks

Power quality issues within the electrical transmission framework can lead to challenges in managing nonlinear loads. Additionally, these power quality risks can result in transient faults and impact renewable generators within the electrical network. Consequently, power electronic components (PEC) are susceptible to these power quality risks. Examples of such risks include voltage swell, voltage sag, and electrical flickers. Furthermore, issues like power surges and unbalanced voltages can adversely affect the performance of equipment, causing fluctuations in voltage, current, and frequency. For instance, harmonic voltage distortion can impact wind energy generation in modern electrical control systems, necessitating balanced sinusoidal current and voltage to maintain consistent magnitude and frequency in the electrical circuit. An overview of power quality disturbance is provided in Figure 3. [1].

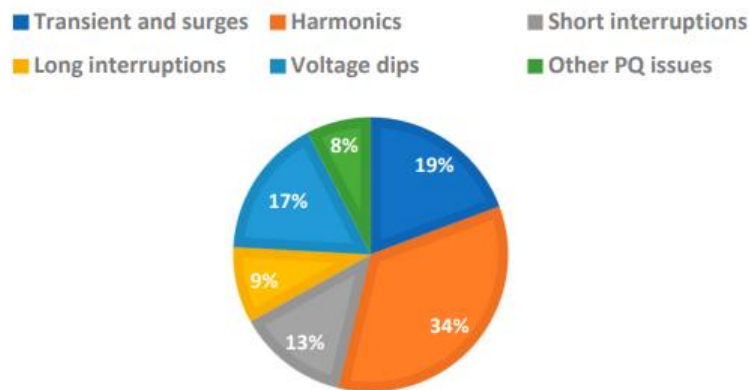


Figure 3: Power Quality Risks

According to Figure 3, harmonics are the predominant factor contributing to substandard power quality, accounting for the highest percentage compared to other risk factors. Transient

and surge risks also significantly impact poor power quality, with the second-highest percentage value following harmonic causes. Overall, the power quality of electric machinery may exhibit imbalanced behavior due to these factors.

Another scenario for this power quality risk is the production of electrical vehicles (EV). The EV market has experienced rapid growth in recent years, driven by increasing environmental awareness, promoting clean transportation solutions and reducing carbon emissions. The advancement in battery technology and storage capabilities of EVs have led to the development of smart charging approaches. This states by reducing charging costs but also helps in balancing the load on the electrical grid. By integrating EVs into the grid through smart charging strategies, grid operators can improve the overall quality, reliability, and sustainability of the grid. However, it is crucial to ensure that the harmonic current content from EV chargers adheres to the limits established by the IEEE Standards. Excessive harmonic currents can have negative effects on the power quality of the electrical grid, leading to voltage imbalances and a decrease in transformer efficiency [15].

Henceforth, the presence of harmonics in AC sinusoidal curves within electrical machinery can be attributed as the primary cause of these power quality risks. Many engineers endeavour to mitigate these harmonics by implementing alternative connections aimed at maintaining electrical power unity and improving quality efficiencies. Methods such as phase balancing, harmonic filters, and series reactance are employed to counteract these negative factors and enhance electricity performance.

On a positive note, I have introduced the artificial intelligence techniques as their presence to address this issue of electrical harmonics in AC sinusoidal waves within electrical circuits. By leveraging AI methods, we aim to analyse, predict, and manage harmonics more effectively, ultimately contributing to improved power quality and grid stability.

1.4. Artificial Intelligence in Electrical Harmonics

Artificial Intelligence (AI) techniques encompass a diverse array of methods and algorithms designed to empower machines with capabilities similar to human intelligence. These capabilities such as perception, reasoning, learning, and decision-making, allow machines to perform tasks. One of the primary objectives of AI method is to analyse complex data, detect patterns within it, and then utilize these intuitions to make informed predictions or decisions [16]. Figure 4 illustrate the behaviour of the AC Sinusoidal wave (Fundamental) and its harmonics.

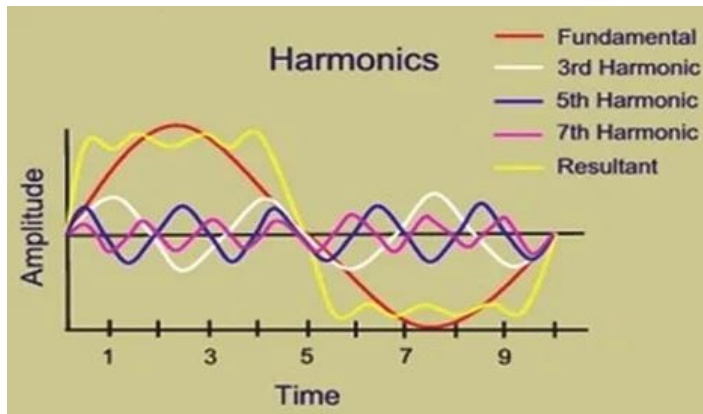


Figure 4: AC Sinusoidal Harmonics

The power quality condition of power electronic equipment and electrical machinery is correlated with this diagram. Harmonics are naturally generated when these AC sinusoidal curves are delivered to equipment devices like AC current and voltage

units, leading to several mistakes. One problem is voltage swell/sag and current transients. The issue of gadgets overheating is another concern. The Equation 1.1.12 illustrates the odd number of Fourier series (b_n) with harmonic behaviour. The fundamental frequency (default $\rightarrow n = 1$) is represented by the colour red. The third harmonic is shown by the colour white. The fifth and seventh harmonic patterns are represented by the colours blue and purple, respectively. As the result, the combination of these harmonic patterns (first, third, fifth, and seventh wavelengths) produces the colour yellow as their resultant index.

Machine learning is a major subset of artificial intelligence. Machine learning approaches involve training algorithms on massive volumes of data, allowing them to detect intricate patterns and relationships. Machine learning algorithms can use this approach to accurately forecast outcomes or make judgments based on the information they have been trained on. This iterative learning process ensures machine learning, as

algorithms continually refine their comprehension and predicting capacities over time [16]**Error! Reference source not found..**

Artificial intelligence has the potential to weaken monitoring, mitigation, and harmonic analysis instruments and methods. In contrast to traditional control, this behavior can be analyzed or predicted into flexible control systems. These techniques attempt to improve harmonic analysis in modern power systems as well [1].

Artificial intelligence techniques face several challenges in complicated real-world circumstances, necessitating sophisticated solutions to successfully manage their tasks and responsibilities. A few factors are:

Load forecasting: Support vector machines, fuzzy logic, and neural networks are examples of artificial intelligence techniques that are used to anticipate electricity demand and help power companies forecast the generation and distribution of power [16].

Fault detection and diagnosis: Errors in the power system may cause long-lasting blackouts, making them hard to locate. To detect and diagnose issues in power systems, artificial intelligence techniques such as neural networks and fuzzy logic expert systems are employed [16].

Power quality assessment: Equipment damage and power outages can result from low power quality when the power system is operating. Wavelet transform, fuzzy logic, and neural networks are examples of artificial intelligence techniques used to assess the power quality of power systems [16].

Smart grid operation: To optimize smart grid performance, artificial intelligence techniques such as reinforcement learning, game theory, and multi-agent systems are used. These techniques balance the supply and demand for electricity, lower energy consumption, and improve power system efficiency [16].

Renewable energy integration: Growing reliance on renewable energy sources, such as wind and solar energy, is making power system management more challenging. Renewable

energy sources are optimized and integrated into the grid by artificial intelligence processes such as neural networks, fuzzy logic, and genetic algorithms [16].

AI methods can be used for spectrum analysis and prediction/estimation of these harmonics' behaviours. The Venn diagram for the AI application (machine learning approaches) in harmonic detection is shown in Figure 5 [1].

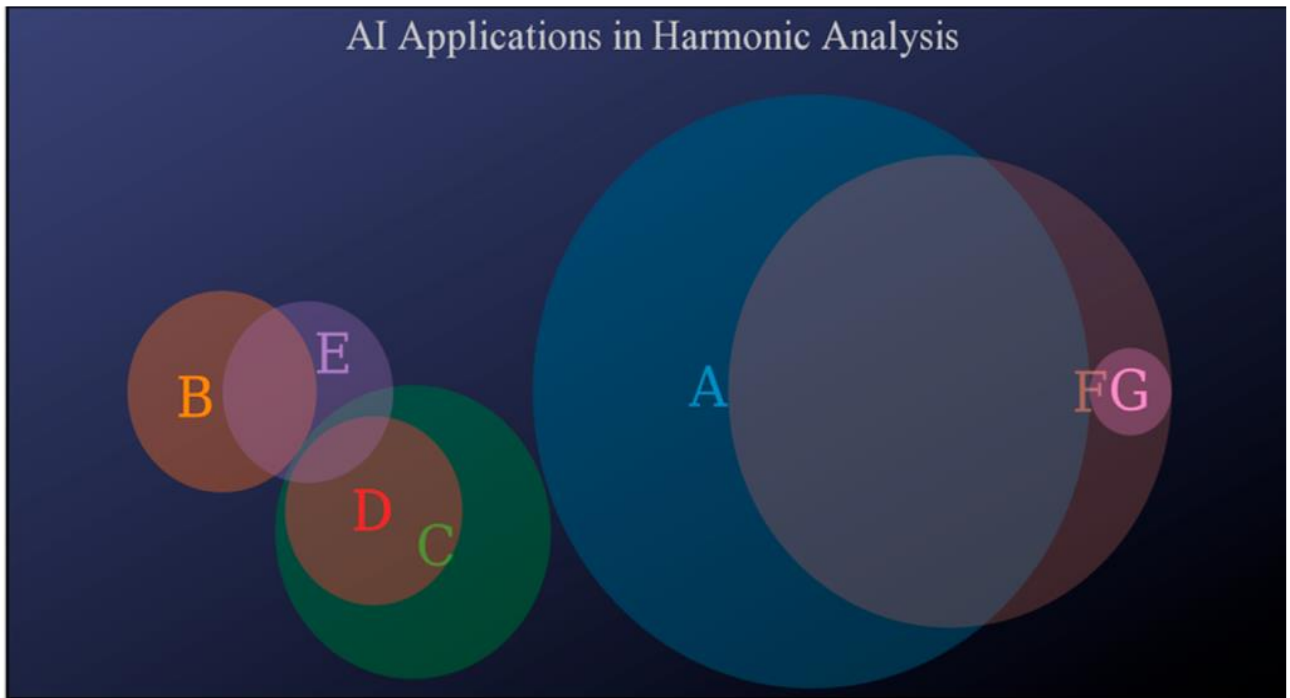


Figure 5: Venn Diagram in Harmonic Analysis

According to above classification:

- A** – Spectrum analysis and estimation/prediction.
- B** – Harmonic source classification.
- C** – Harmonic source location and estimation.
- D** – Harmonic source contribution.
- E** – Harmonic data clustering.
- F** – Harmonic elimination and Active Power Filter.
- G** – Distributed Generation hosting capacity harmonics [1] .

Agricultural operations, for instance, are currently using artificial intelligence with robotic technology. The most advanced AI-based farming system is replacing the traditional agricultural system. In this system, the computer itself takes decisions for problem-solving in real time (Machine Learning Topic). Currently, scientists and engineers are putting forth a lot of effort to combine these agricultural aspects into a system that is

straightforward, intelligent, affordable, highly productive, time-efficient, sustainable, and that will support a prosperous and healthy farming society. On the other hand, robotics and other part of machineries are relevant nowadays. They are found in complex processes, sensors, data management, the Internet of Things (IoT), and computers' algorithms that make intelligent decisions as accuracy [17].

Nowadays, artificial intelligence is widely utilized in numerous fields and can operate quickly. This is the result of the continuous development and optimization of machine learning algorithms to handle a variety of problems, including those in the healthcare sector. The application of AI to medical imaging is therefore one of the most important fields of study. To achieve credibility, causation, transferability, confidence, equity, accessibility, and interactivity, Experience AI (XAI) is an explainable model that generates predictions. This allows the public to understand the model's decision-making in the healthcare industry [18].

Countries will experience a water crisis nowadays. As a result, these crises have a logic of the world's dense population, erratic weather patterns, inadequate management of water resources, and natural calamities. There are significant issues caused by solid waste and litter dumped in bodies of water. Hence, there are alternative application of AI technologies in this field that can address the efficient management of water resources, building of efficient supply systems, optimization of current water resources, and development of infrastructure for water resources and safe water mechanisms. Artificial intelligence in the water supply has policy implications to enhance water utility efficiency and free up high-quality services. These applications of AI greatly enhance the general state of environmental quality. Water is therefore one of the most essential resources for daily life, demanding advanced management strategies [19].

1.5. Artificial Intelligence Techniques Advanced

Techniques in artificial intelligence encompass a wide range of approaches and algorithms. These components might enable machines to do intelligent tasks that resemble human intellect and are sustainable. Among these tasks are perception and decision-making too. This methodology also makes use of algorithms to correlate computer vision, robotics,

natural language processing, and machine learning. It deciphers complex data, looks for patterns, and then forecasts or predicts using those patterns. The field of artificial intelligence known as "machine learning" focuses on programming algorithms to identify patterns in data so they can forecast or make predictions. The following lists are a few machine learning algorithms.

Neural Networks – referred to as artificial neural networks (ANNs) are node-based computational models. This system uses mathematical calculations to process information and is layered in structure. Based on facts, this network is able to forecast or form opinions. Consequently, it might use the neuronal weights and biases to select the training process. Reducing the difference between the network's real and expected output is the aim of this character [20]. ANNs can therefore be programmed to recognize AC sinusoidal harmonics. The neural network notion in mathematics is depicted in Figure 6.

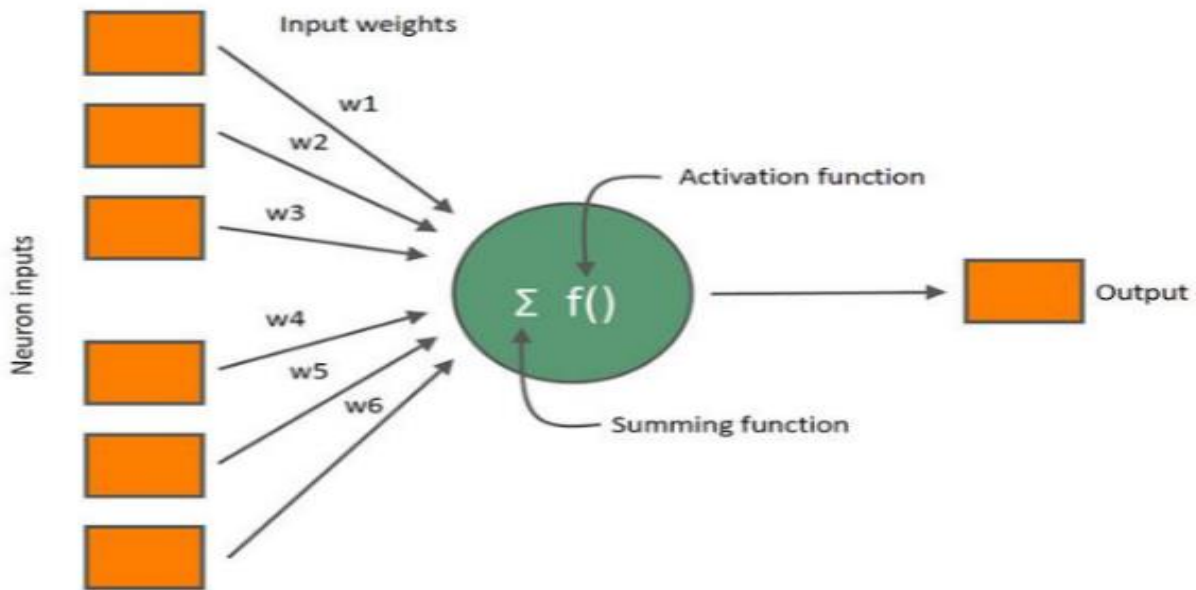


Figure 6: Neutral Network Mathematic Algorithms

- **Neuron Input (Orange Color)** – represent the inputs to the neuron.
- **$w_1, w_2, w_3, w_4, w_5, w_6$** – Weights determine the strength and direction (positive or negative) of the influence of inputs

- **Summing Function (Σ)** – all weighted inputs are summed together represented and then passed through an activation function ‘f’.
- **Output (Orange Color)** – indicates the final output of the neuron after processing by the activation function.

Fuzzy System – a kind of artificial intelligence that is widely used between a system's inputs and outputs in electrical power grid applications. It is possible to express incomplete truths quantitatively. Furthermore, this technology has the potential to enhance energy efficiency, facilitate improved grid management, and elevate the standard of power supply. It also controls generators, transmission lines, and distribution networks by reducing voltage instability, frequency variations, and power quality monitoring. Power quality disturbances, such as voltage sags, swells, and harmonics on sensitive equipment, can be recognized and categorized by it. Fuzzy systems have also been applied to demand-side management in order to optimize energy consumption, reduce peak demand, and forecast energy consumption trends to cut-down energy usage at peak times [16]. Figure 7 depicts the Fuzzy System structure.

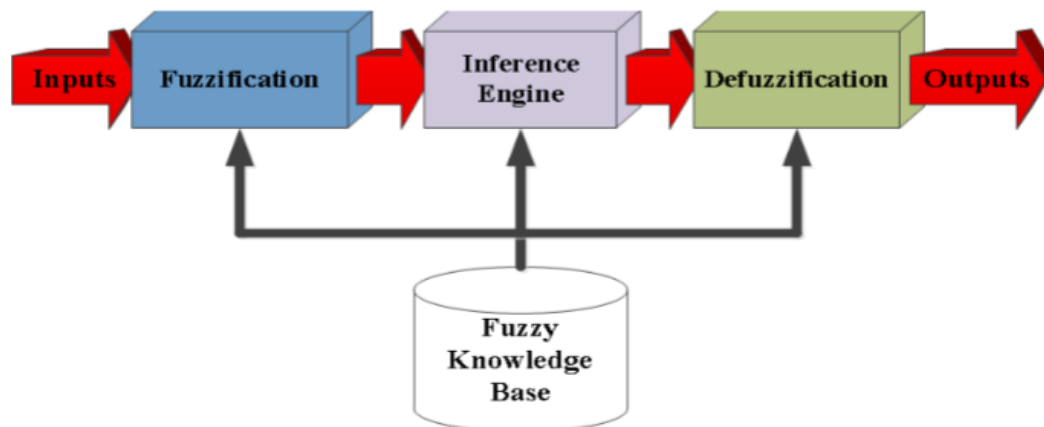


Figure 7: Fuzzy Algorithms System

Inputs – represent the initial fed into the fuzzy logic system. An example is temperature control system, the input could be the current room temperature.

Fuzzification – Converts inputs into fuzzy sets (degree to each data input).

Inference Engine – Applies logical operations to fuzzification inputs and derives fuzzy output sets.

Defuzzification – Converts fuzzy output sets into values.

Outputs – Final values derived from processing fuzzified inputs through inference and defuzzification.

Support Vector Machines (SVMs) – a type of machine learning method that maximizes class separation for regression analysis and classification by identifying the optimal hyperplane in a high-dimensional feature space. This algorithm generates a hyperplane(s) with n properties in an n -dimensional space. The goal is to maximize the margin between the hyperplane and the closest data points from each class. Using information gathered from the power stream, this machine learning approach identifies various problem types in the power grid and correlates the fault diagnosis. It can be trained to identify whether a defect is single-phase-to-ground, phase-to-phase, or three-phase based on the signal characteristics extracted from the power signal. In a similar case, attempts have been made

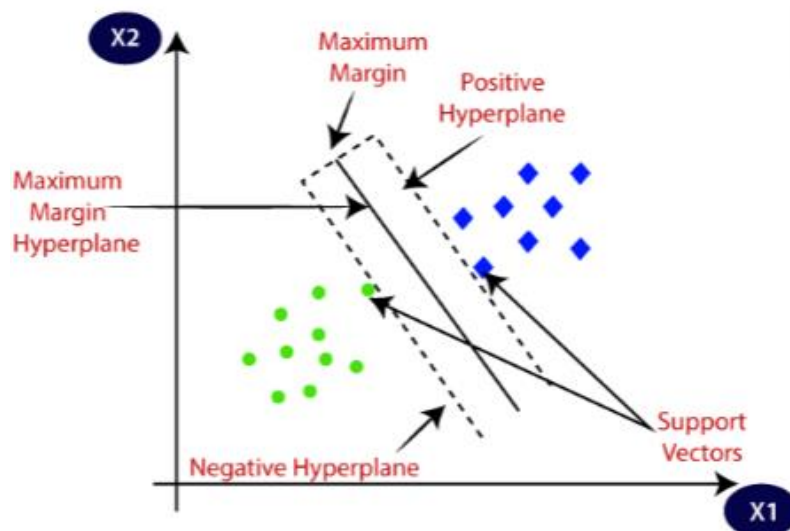


Figure 8: Support Vector Machine's Hyperplane

to categorize and forecast power system faults as well as forecast future electricity consumption [16]. Figure 8 displays the SVM method in a plane graph.

1.6. Summary

Chapter One provides an overview of AC Sinusoidal Harmonics, including explanations of fundamental and harmonic frequencies, illustrated through images, and a discussion on the Fourier Mathematical Series. Additionally, this chapter investigates into the impacts faced by electrical machinery and power electronics due to inadvertent harmonics effects in AC Sinusoidal curves, leading to heightened risks to power quality and increased performance costs, such as overheating temperatures. Consequently, the chapter explores the role of Artificial Intelligence in detecting and predicting these harmonic effects, offering insights into machine learning principles and algorithms.

CHAPTER 2: LITERATURE REVIEW

2.1. Literature Impact

My research thesis examines the exploration of literature concerning harmonics in AC sinusoidal waves, power quality, and artificial intelligence techniques. Within the vast array of available articles, I meticulously curated a subset of significant publications, each shedding light on the multi-dimensional definitions and applications of these factors across various modern contexts. For further clarification, readers are encouraged to refer to the reference section of my thesis.

This thesis synthesizes elements from both power engineering, which serves as the focal point of my research analysis, and computer science, even though in a minor capacity. Within the domain of computer science, emphasis is placed on algorithms, computer architecture, and computation methods across various academic disciplines. In the dominion of power quality, scholarly materials investigate electrical harmonics within AC sinusoidal waves, encompassing both voltage and current units. Relevant topics include Fourier series, harmonic distorted waveform lengths, with a particular emphasis on the modulation of harmonic cycles through odd-order harmonics (Fourier Series: b_n).

My exploration of research articles has primarily centered around voltage harmonics wavelengths in smart grids, coupled with the utilization of artificial intelligence and machine learning techniques at their core. Among these articles, there is a surplus of methodologies focused on machine learning techniques within the realm of computer science. Currently, I am discerning options that align with the essence of my research thesis. Interestingly, I have encountered several articles that explore the application of machine learning techniques in the analysis and interpretation of harmonic factors.

2.2. Literatures of Power Quality Risks

The definition, traits, and causes of harmonics in power systems, including nonlinear loads, power electronic converters, and renewable energy sources, are covered in detail on the SEA Energy research website, "Harmonics in Electrical Power System". It ascertains how harmonics affect equipment performance, power quality, overheating, and interference with delicate electronic equipment. This study also demonstrates technologies and techniques that cause harmonic analysis, measurement, and modeling in power systems. These include active and passive filters as well as harmonic cancellation strategies [11].

Khaleel et al express that power quality significantly impacts the competitiveness of various activities. Expenses related to power quality problems can affect outcomes. These issues are exacerbated by the growing number of customers affected by PQ problems. Financial costs associated with PQ problems include production downtime, defective products, and indirect expenses. According to the Institute of Electrical and Electronics Engineers (IEEE), supply quality encompasses power provision and grounding for electronic devices, ensuring their suitability for operation and compatibility with the wiring system and other devices. This issue has gained importance for both energy distribution companies and end-users due to various factors [16].

In-depth examination of harmonic-related issues in power systems, including estimate methods and harmonic distortion, sources, effects, and electrical engineering significance, are explained by Taghvaie et al. They go over the fundamentals, benefits, and drawbacks of various harmonic estimating and analysis techniques, including the Fourier and Wavelet transforms [5].

2.3. Literature of Proposed Harmonic System

However, Mahmoud et al. proposed the proof that sounds as a more viable approach compared to the conventional strategy of utilizing 180-degree stage move in electrical loads. They advocate for the execution of dynamic control channels, such as shunt, arrangement, and shunt-plus Inactive Synchronous Compensator (STATCOM). These factors together can control the Unified Power Quality Conditioner (UPQC), over detached channels. This approach points to significantly decrease non-linear stack currents inside AC Sinusoidal bends. STATCOM strategies offer enhancements in power quality by improving cutting edge electrical control framework steadiness and successfully directing voltage sag and swell through sinusoidal pulse width modulation [21]. This proposition is to some degree comparative concept to my research thesis methodology.

2.4. Reviews of Artificial Intelligence factor

Ugwuagbo et al. highlighted the significance of artificial intelligence techniques, such as neural networks and support vector machines, in handling synthetic data for power system classification. This specific data plays an active role in classification methods, particularly in single-phase power quality scenarios [12]. Therefore, the utilization of AI techniques becomes crucial, especially in cases where harmonic AC Sinusoidal curves are present in power quality loads.

In addition, this study emphasizes the value of real-time data acquisition and processing in enhancing the responsiveness of voltage instability detection systems with integration of machine learning techniques. Mehinović et al. provide a thorough analysis of the application of advanced machine learning techniques in power systems, detected voltage instability issues with algorithms and their effectiveness in predicting voltage instability events, such as support vector machines, artificial neural networks, and decision trees. They clearly stated these voltage instability challenges in which importance of feature selection and data preprocessing techniques and in improving the accuracy of voltage instability prediction models when performance of machine learning algorithms [3].

A swift of AI and ML methods and their growing use in power system analysis and monitoring is also provided by Taghvaie et al. The topics covered relate to current developments in deep learning models and harmonic estimating techniques in AI and ML. It looks for the benefits that these strategies have having over more than conventional approaches, including increased automation, flexibility, and accuracy. Additionally, this study examines case studies in which conventional and AI/ML-based methods have been used to handle harmonic problems in power systems, raising potential difficulties and unanswered research concerns about harmonic estimates [5].

2.5. Summary

Chapter two provides an overview of AC Sinusoidal Harmonics, including explanations of fundamental and harmonic frequencies, illustrated through images, and a discussion on the Fourier Mathematical Series. Additionally, this chapter investigates into the impacts faced by electrical machinery and power electronics due to inadvertent harmonics effects in AC Sinusoidal curves, leading to intensified risks to power quality and increased performance costs, such as overheating burst of temperatures. Consequently, this chapter explores the role of Artificial Intelligence (AI) in detecting and predicting these harmonic effects, offering insights into machine learning principles and algorithms.

CHAPTER 3: SIMULATION METHODOLOGY

My Master's Degree research thesis methodology covers defining simulation and evaluating its performance. In addition, various hardware materials are utilized for electrical harmonic testing. Solar Photovoltaic electrical circuits are evaluated for harmonic behavior when operating with renewable energy production. Another component of studying harmonic behaviors on hardware materials is the voltage converter model, which operates from AC to DC in a manner similar to rectifier electrical circuitry. Figures 9 and 10 are presented in the Power and Smart Grid Laboratory - Room 1. However, I must follow standards by first simulating harmonic behavior before proceeding with hardware access. As a result, my thesis includes both software and hardware components; nevertheless, my primary focus is on software for experimental testing settings. The abundance of research resources for computer analysis and its algorithms is the cause of this thesis. The focus of my research thesis, which serves as evidence, is only on the use of artificial intelligence and machine learning techniques for the detection of voltage harmonics within AC sinusoidal waves.



Figure 9: Solar PV Harmonic Circuit



Figure 10: Harmonics in Converter Model

3.1. Detecting Harmonics using Primary Simulation Software

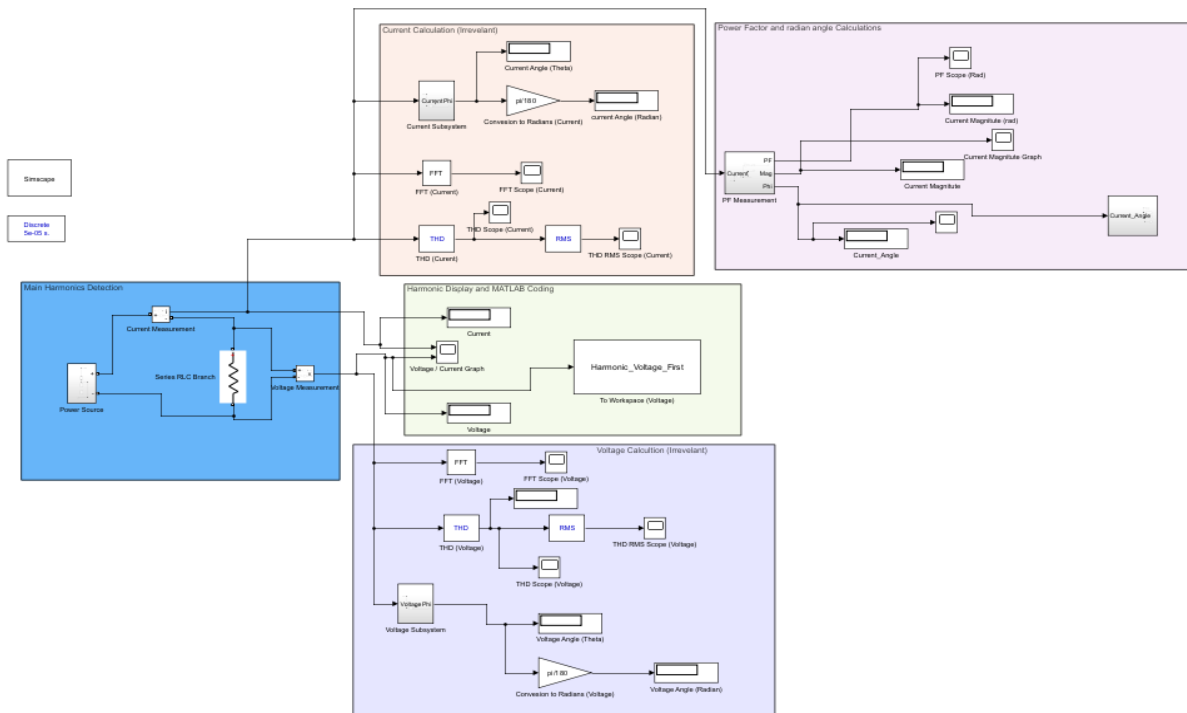


Figure 11: MATLAB Simulink for Harmonics

The investigation of my thesis integrates aspects of Power Engineering with data retrieval techniques from Computer Science. I constructed a fundamental simulation electrical circuit with sinusoidal harmonic waves using MATLAB Simulink software. Figure 11 shows the MATLAB Software, separated in colors. This diagram will be used in chapter 4 (Results and Discussion).

Figure 11 illustrates the primary harmonic producer (depicted in blue), which generates multiple AC sinusoidal waves through the AC Synchronized Waves block within the Power Source. This color block can produce sinusoidal waves with frequency, phase, and amplitude variations. Also, this function can produce more than one sinusoidal wave, like electrical harmonics. These waves then traverse through the series RLC component as part of the electrical circuit. The Voltage and Current Management components are responsible for detecting these waveforms. Subsequently, the waveform data is transmitted to the Harmonic Display and MATLAB Coding blocks (depicted in green). The green block's main application is the workspace block where these waves can be easily detected and retrieve it into the plotted data points. The MATLAB Coding for detect these harmonic Data are shown below

First Harmonic Data:

```
% Assuming Voltage_First is your simulation output

Time = Harmonic_Voltage_First.Time; % Access the time data

Data = Harmonic_Voltage_First.Data; % Access the voltage data

T = table(Time, Data); % Create a table

writetable(T, 'harmonicFirst.csv'); % Write to CSV
```

First and Third Harmonic Data:

```
% Assuming Voltage_First_Third is your simulation output

Time = Harmonic_Voltage_First_Third.Time; % Access the time data

Data = Harmonic_Voltage_First_Third.Data; % Access the voltage data

T = table(Time, Data); % Create a table

writetable (T, 'harmonicFirstThird.csv'); % Write to CSV
```

First, Third and Fifth Harmonic Data:

```
% Assuming Voltage_First_Third_Fifth is your simulation output

Time = Harmonic_Voltage_First_Third_Fifth.Time; % Access the time data

Data = Harmonic_Voltage_First_Third_Fifth.Data; % Access the voltage data

T = table(Time, Data); % Create a table

writetable (T, 'harmonicFirstThirdFifth.csv'); % Write to CSV
```

3.2. MATLAB Simulink Assessment

While investigating research articles for my proposal, I simultaneously proceeded to the implementation stage. I found that the MATLAB program was the ideal tool for analyzing harmonic behaviors, leading me to utilize MATLAB program in SIMULINK mode. Within the SIMULINK environment, I employed the AC Synchronized Waves block in Simscape mode to simulate the generation of harmonic wave outputs. Another essential feature is the powerGUI function, enabling simulation within SIMULINK and

Power System functionalities in this circuit. This approach facilitated software-based experimentation, allowing for the generation and manipulation of harmonic waves, encompassing both voltage and current units. Despite the graphical representation of these measurements on a unified scale, achieving precision and accuracy in capturing harmonic wavelengths remains a priority for improvement.

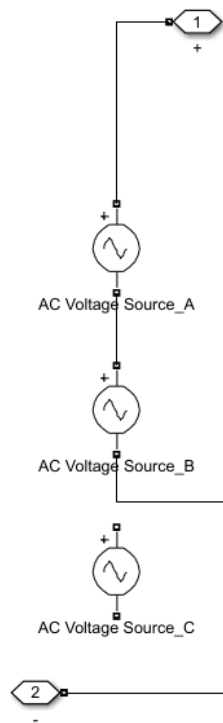


Figure 12: Power Simulation Source

The design process involved using three AC Sinusoidal blocks in this SIMULINK program, each configured with varying frequencies, amplitudes, and phases to generate distinct harmonic patterns. This setup closely mirrors real-world AC waveforms, which forms the core focus of my research proposal. Figure 12 illustrates MATLAB's Power Source.

Furthermore, I utilized the voltage measurement block in MATLAB SIMULINK to extract AC Voltage from the controlled AC Sine wave pattern, enabling visualization through voltage waveform plots. Although the option to introduce loads such as nonlinear applications or smart grid circuits existed, I deemed it unnecessary for the primary objective of identifying harmonic waves.

In addition to voltage measurements, the current measurement block was used to analyze AC sine wave currents. Further advancements are made by incorporating additional blocks in the SIMULINK environment, including power factor detection using current magnitude and phase measurements in radians. However, challenges were encountered in plotting FFT (Fast Fourier Transform) and THD (Total Harmonic Distortion) charts, requiring further refinement in this aspect. Therefore, voltage measurements are pivotal for the conclusion of this research proposal experiment thus far.

3.3. Regatron ACS Machine Assessment

The second approach, rather than solely focusing on MATLAB SIMULINK program, involves utilizing the Regatron TopCon ACS Machine. This device is an integral part of power electronic sources, acting as a converter that enables a power amplifier to generate harmonic testing from the main power grid system. It is specialized equipment designed for testing and simulation purposes across various industries. This system simulator is capable of designing and simulating electrical system conditions in both grid-feeding and grid-absorbing modes. It can operate in all four quadrants of the power factor space, allowing for comprehensive testing of power electronic converters, renewable energy systems, and other electrical devices under different system conditions. This machine serves as a hardware component for testing harmonic behaviors in solar PV circuits and current conversion converter machines for classification. Figure 13 illustrates the hardware machine at the Power and Smart Grid Laboratory.

Consequently, I conducted a research experiment on this Regatron TopCon ACS machine, a Power Amplifier, and explored its Smart Grid simulation capabilities using LabVIEW software as part of additional equipment research. I may consider utilizing this harmonic data in the Automation department under the LABVIEW program if time permits in my Master Research proposal. Nonetheless, I generated harmonic waves using the Regatron ACS program. Typically, I must generate these voltage units first with odd-number multiples (3x, 5x, etc.) to incorporate into the Harmonics mode.



Figure 13: Regatron ACS Machine

However, there is a limitation in the time range for storing this AC sinusoidal wave data. Storing data for a specific duration, such as 10 seconds, requires a significant amount of time if done through external methods. This may position a challenge for Machine Learning (ML) models that require extensive data for prediction. Despite this limitation, I successfully managed to store the harmonic data in CSV format multiple times (5x, 10x), obtaining similar data patterns for further analysis.

3.4. Jupyter Notebook Evaluation

Now, I possess both software (MATLAB) and hardware (Regatron TopCon) methods for acquiring these harmonics data. The next step involves testing these datasets using Machine Learning techniques. The methodology applied to these datasets can encompass various predictions and forecasts of harmonics in the future time-frames. Jupyter Notebook software, apart from renewable Energy Methodologies proves to be an ideal tool for analyzing the issues present in this data regarding on electric harmonics; a key for power quality improvement.



Figure 14: Jupyter Notebook Solutions

Jupyter Notebooks are increasingly used due to their simplicity, the support community, the availability of libraries, and because they are open educational resources and support most popular programming

languages. It is self-regulated learning that has been gaining ground due to its interactive environment and makes it easy to manipulate codes and adjust their parameters.

Programming in Python language is well-suited for implementing the Jupyter Notebook and extracting answers from the harmonic data values. It is essential to note that the data, however, is not entirely clean, due to the hardware system from the Regatron TopCon machine; a disadvantage rather than pure AC Sinusoidal Data. As a result, I

employed another AI software called Orange Data Mining. This software can clean these harmonic data to the accuracy bench as well as machine learning data and visualization too (See Figure 15).

3.5. Orange Data Mining Analyst

Orange Data Mining software is an AI software compromise with machine learning techniques. This software can preprocess the data as well to cleanse conditional data too. Therefore, I replicated the same procedure for my harmonic data, aiming to achieve cleanliness acquired data. Orange Data software instead of cleaning data has several Machine Learning techniques. A few are employed prediction algorithms, K-Nearest

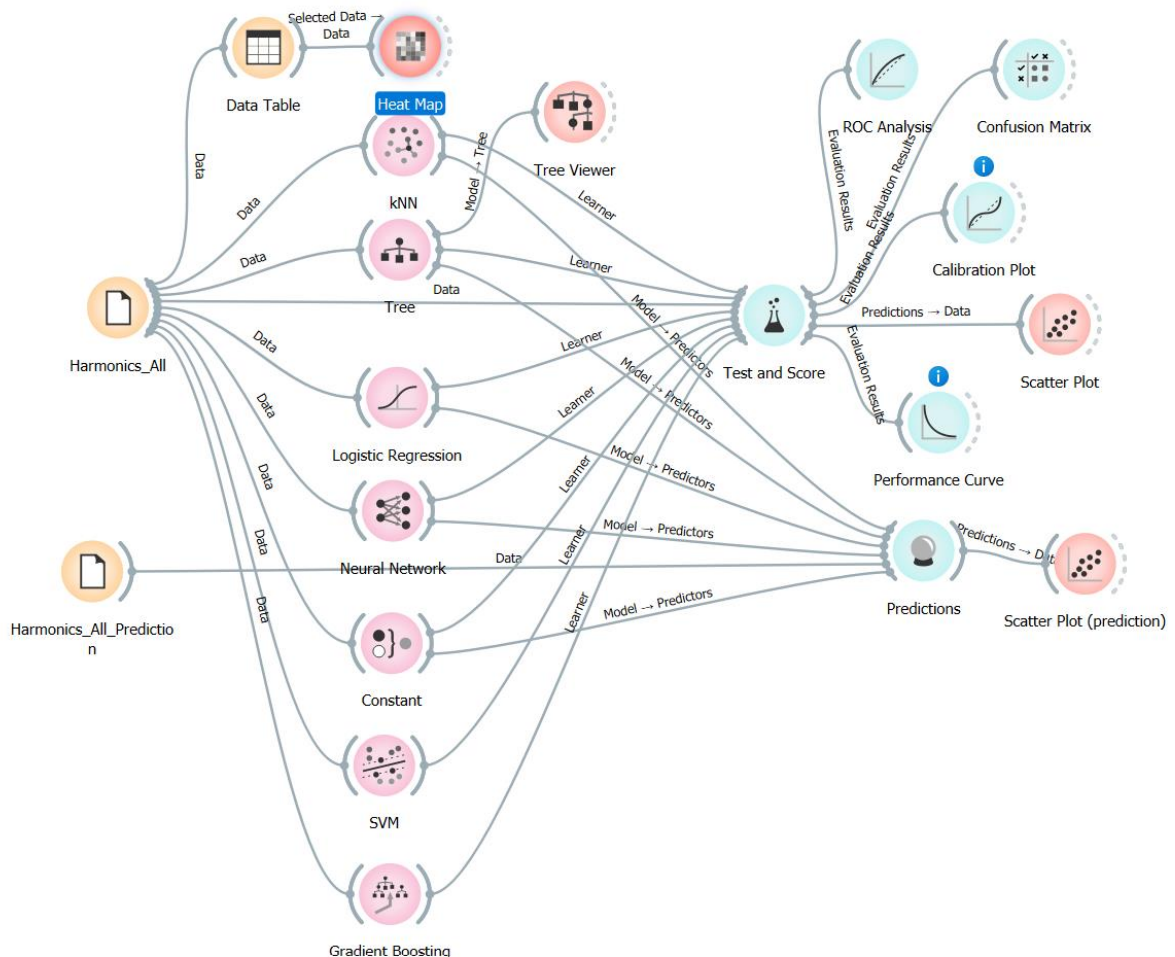


Figure 15: Orange Data Mining AI Software

Neighbors (KNN), Random Forest and Support Vector Machine techniques. In Figure 15, the Orange data with harmonic csv data from both software and hardware's analyst is depicted above. A few Machine Learning Models are further explained below after the figure 15.

K-Nearest Neighbors (KNN): Used for classification and regression tasks of AC sinusoidal harmonics. It classifies different harmonic patterns based on their features extracted from the waveform data.

Decision Tree (DT): Hierarchical models that partition the feature space into regions based on simple rules. Decision trees can identify characteristic patterns in AC sinusoidal harmonics waveform data, leading to classification or regression tasks.

Logistic Regression (LR): A statistical method used for binary classification tasks. In AC sinusoidal harmonics, logistic regression classifies whether a waveform exhibits harmonic distortion based on certain features extracted from the waveform data.

Neural Network (NN): Particularly deep learning models capable of learning complex patterns from data. In the case of AC sinusoidal harmonics, neural networks automatically extract features from waveform data and predict the presence or characteristics of harmonics.

Support Vector Machine (SVM): A supervised learning algorithm used for classification and regression tasks. SVM classifies several types of AC sinusoidal harmonics based on the separation of feature vectors extracted from the waveform data.

Gradient Boosting (GB): An ensemble learning technique that builds a strong predictive model of AC sinusoidal harmonics. It iteratively improves the prediction of harmonic patterns by learning from the errors of previous models.

This software closely aligns with Machine Learning techniques, proving to be a valuable tool for ongoing research thesis. Data cleaning involves removing excessive points located away from AC sinusoidal points. To ensure clean harmonic data, several sets of data instead of one, focusing on quality rather than quantity—an approach conducive to effective AI and ML data prediction. While a small amount of data may not yield high accuracy and predictive points, numerous points or big data can increase accuracy output and overall efficiency. Hence, ten sets of CSV data from the Regatron Machine and also MATLAB SIMULINK software, each programmed to represent the first harmonic and first and third (odd numbers) harmonic AC sinusoidal curves. Subsequently, filtered out unnecessary points using the Orange data software, generating clean data resembling a pure AC sinusoidal curve with harmonic factors.

The next step in the analysis is to examine the test and score data using the Orange Data Mining software. This block displays the accuracy and performance of the harmonic curve points. Below is the list of the test and score analyses.

AUC – Area Under the ROC Curve measures the area under the receiver operating characteristic (ROC) curve, which plots the true positive rate against the false positive rate [22].

CA – Classification Accuracy measures the proportion of correctly classified instances out of the total instances indicates the overall correctness of the model's predictions [22].

Recall – sometimes called Sensitivity or True Positive Rate measures the proportion of true positive instances that are correctly identified by the model out of all actual positive instances [22].

Precision – called Positive Predictive Value measures the proportion of true positive instances among all instances predicted as positive by the model and to avoid false positives [22].

F1 – named F1 score is the harmonic mean of precision and recall. Calculated as $2 \times (Precision \times Recall / (Precision + Recall))$ and provides a balance between precision and recall where there is an imbalance between the classes [22].

MCC – Matthews Correlation Coefficient is a correlation coefficient that measures the quality of binary classifications, takes true and false positives and negatives and ranges from -1 to 1, where 1 indicates perfect predictions, 0 indicates random predictions, and -1 indicates inverse predictions. It only produces a high score in its range (-1: +1) if the classifier achieved high scores for each of the confusion matrix's four fundamental rates: sensitivity, specificity, accuracy, and negative predictive value [23].

Spec – called Specificity measure of the model's ability to correctly identify true negative instances out of all actual negative instances. Represents the proportion of true negatives (correctly identified negative instances) over the total number of actual negative instances. A higher specificity indicates that the model is better at correctly identifying negative cases. It also determines such applications across domains that can affect competitive advantage like Chatbot. The applications are classified into four types of machine learning: generic, distinctive, selective, and exclusive [24].

LogLoss – Logarithmic Loss or cross-entropy loss, measures the performance of a classification model where the prediction output is a probability value between 0 and 1, quantifies the accuracy of the predicted probabilities compared to the actual class labels. It is also logarithmic transformation of this function and penalizes incorrect classifications based on the difference between predicted probabilities and actual outcomes. For instance, a lower cross-entropy loss value indicates better performance, with perfect predictions resulting of zero [25].

This topology outlines each stage of research case study. Initially, focusing on harmonic AC Sinusoidal wavelengths and their analysis using artificial intelligence techniques. The next step involved designing methods to produce harmonic wavelengths, utilizing both software and hardware approaches (Referring to bullet points after figure 16). Specifically, two methods as outlined: MATLAB SIMULINK software and hardware implementation using the Regatron TopCon ACS Machine (Repeat Case). Subsequently, the generated harmonic data, including voltage units, was saved as CSV files. These files were then imported into artificial intelligence and machine learning techniques, as Jupyter Notebook AI software and Orange Data Mining ML Software. These software tools are instrumental in analyzing the results, making predictions, and forecasting harmonic patterns. Additionally, they provided metrics such as accuracies, precisions, and F1-score data, along with machine learning models and visualizations, facilitating the control of harmonic patterns in advance. These evaluations determine whether the test accuracy aligns with the patterns and values observed in the harmonic AC sinusoidal curves. This fundamental concept supports the essence of my research experiment.

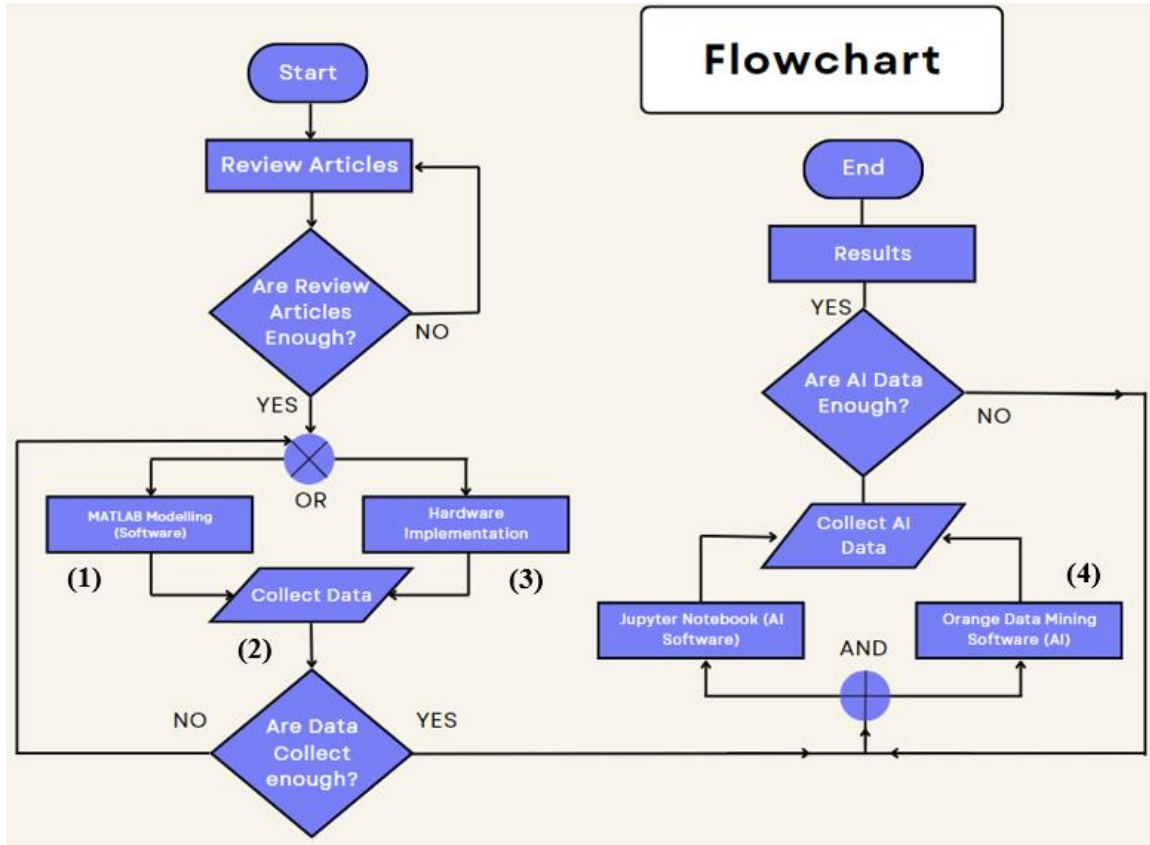


Figure 16: Flowchart for Analyst Harmonics

A few of steps and discussions in the figure 16 are listed below:

Produce MATLAB SIMULINK harmonics (Ref: (1) above diagram: fig 16): The MATLAB SIMULINK electrical circuit (see Figure 10) is designed to produce harmonic patterns according to the power source. This is achieved by utilizing the power source block with variable frequencies, phases, and amplitudes for voltage phasors. Moreover, visualization is shown to detect the harmonic curves from the power source.

Obtain MATLAB SIMULINK data (Ref: (2)): After discovering the harmonic patterns from the power source, the key factor is to capture these waves and store them. The Workplace block is utilized to capture these AC sinusoidal curve points (Time and Voltage amplitude). Initially, voltage is identified. Next, MATLAB Script algorithms are

programmed to run the codes (Green Color on MATLAB Figure 10) until access to the data sheet (CSV) with both time and voltage points is obtained.

Hardware Implementation (Ref: (3)): The Regatron TopCon ACS Machine or any other machines capable of producing harmonic waves are accessible for this implementation. The harmonic odd-numbered orders are designed by monitoring the Regatron Machine's visualization monitor. Initially, only the first and first-third orders are tested, with 100V and 5V respectively. Figures of Regatron TopCon ACS Machine are shown in chapter 5 (Results and Discussion).

Hardware Implementation extract (Ref: (2)): The machine-produced harmonic waves can be observed in the electrical behavior transferred to an AC sinusoidal oscilloscope machine as the output version. This output can extract harmonic points too.

Orange Data Mining Cleansing: The harmonic data from the Sinusoidal oscilloscope of the Regatron Machine is not pure sinusoidal curves due to the inferior quality of hardware detected. Therefore, unwanted points can be voided using the Orange Data cleansing. This software has the principle of Machine Learning and hence can be used with its models too.

Orange Data Mining Detect: Having both software and hardware data on pure AC harmonics, the Orange Data software can utilize this data to implement machine learning models and generate visualizations. As depicted in Figure 16, the Test and Score block showcases the outputs of these models (Refer Chapter 4 – Discussion and Results).

CHAPTER 4: RESULTS AND DISCUSSIONS

For my Master Research thesis, I plan to leverage both MATLAB SIMULINK and the Regatron Machine due to their capabilities in handling harmonic data. MATLAB SIMULINK serves as an ideal virtual integrator, allowing for the generation of harmonic AC sinusoidal waves with flexibility in adjusting parameters such as frequency, amplitude, and phase in Electrical Circuit's characteristics.

In my MATLAB SIMULINK setup, I have devised a system comprising three AC sinusoidal blocks, collectively referred to as "Harmonic Power Source." These blocks enable the creation of diverse harmonic patterns by manipulating the input parameters. For instance, I can configure the first block to produce a sinusoidal wave at 100 V and 50 Hz, maintaining consistency across the system. Meanwhile, I vary the parameters of the subsequent blocks, such as setting the third block to generate a wave at 5 V and 70 Hz. Consequently, I can use 02 sinusoidal blocks as depicted in figure 17 below or else I could penetrate with these three sinusoidal waves too.

It is important that I specifically focus on utilizing odd-numbered harmonics for visualization purposes. By doing so, I aim to capture distinctive patterns inherent in harmonic waves, particularly when they assume a curve shape. While even-numbered harmonics could also be incorporated, their contribution tends to yield smaller values, thus offering negligible insights in comparison.

Through this approach, I anticipate gaining valuable insights into the behavior and performance of these harmonic components within the system, contributing to a deeper understanding of the underlying phenomena.

The parameters examined in this study have presented numerous technical challenges when analyzing the harmonics of AC Sinusoidal waves using artificial intelligence methodologies. Just to inform you, I can vary the power voltage sinusoidal more than once, indicating that harmonics can manifest at various stages with varying amplitudes and frequencies as well.

4.1. MATLAB Simulink Simulation AC Sinusoidal Graph Results

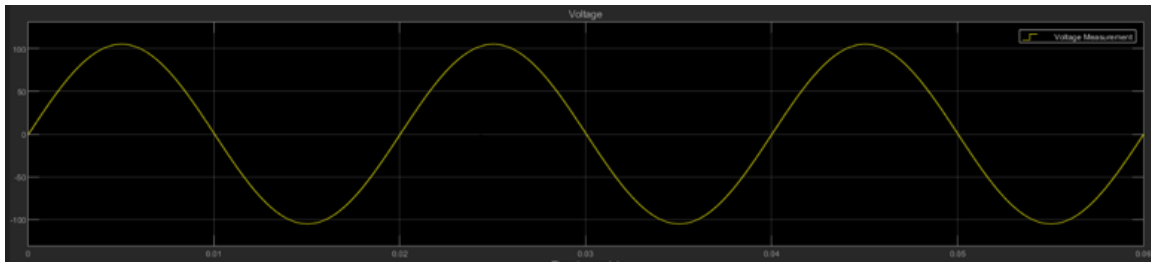


Figure 17: Simple Harmonic Waves (Fundamental)

Figure 17 Harmonic Details: First AC Sinusoidal Voltage – 100V (Amplitude), 50 Hz

Third AC Sinusoidal Voltage – 5V (Amplitude), 70 Hz

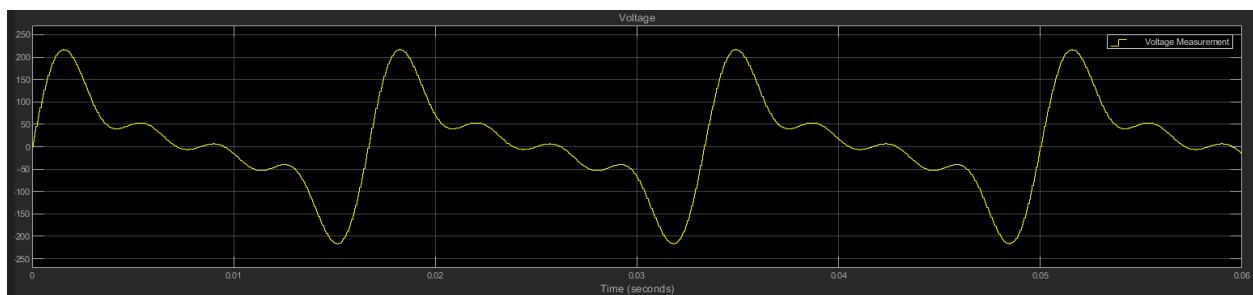


Figure 18: Vary Harmonic Waves

Figure 18 Harmonic Details: First AC Sinusoidal Voltage – 100V (Amplitude), 50 Hz

Third AC Sinusoidal Voltage – 80V (Amplitude), 100 Hz

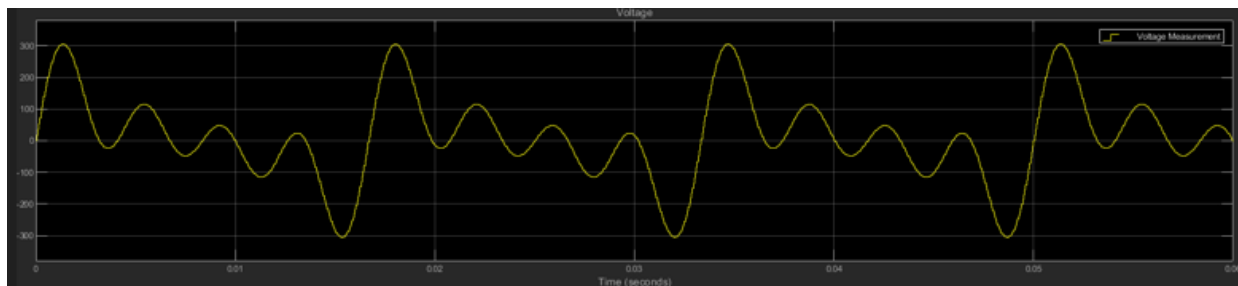


Figure 19: Three AC Sinusoidal Curve

Figure 19 Harmonic Details: First AC Sinusoidal Voltage – 100V (Amplitude), 50 Hz

Third AC Sinusoidal Voltage – 80V (Amplitude), 100 Hz

Fifth AC Sinusoidal Voltage – 60V (Amplitude), 150 Hz

To provide improved accuracy, data from both software and hardware sources must be of sufficient volume. The 'Big Data' course provides significant insights into working with enormous datasets. This dataset is essential for evaluating harmonic waves and power quality variables with machine learning algorithms.

According to the Orange Data mining results, I have three clean datasets with time and voltage values, denoted by the indexes 'zero, one, and two'. These datasets are collected from MATLAB Simulink Harmonics using the Orange Data Mining environment.

4.2. MATLAB “Zero” Harmonic on Orange Data Mining Results

Table 1: Harmonic “Zero” Test Score Curve

Model	AUC	CA	F1	Precision	Recall	MCC	Spec	Log Loss
kNN	0.995	0.971	0.956	0.961	0.952	0.934	0.980	0.109
Tree	0.773	0.581	0.613	0.443	0.994	0.401	0.375	0.461
Logistic Regression	0.469	0.551	0.296	0.310	0.284	-0.033	0.684	0.637
Neural Network	0.521	0.470	0.495	0.363	0.779	0.099	0.316	0.618
Constant	0.500	0.600	0.250	0.332	0.200	-0.001	0.799	0.637
SVM	0.491	0.496	0.435	0.347	0.582	0.033	0.453	0.639
Gradient Boosting	0.710	0.562	0.577	0.425	0.897	0.300	0.394	0.526

According to KNN features in Table 1 for current harmonic test data, the area under the harmonic ‘zero’ curve is 99.5%, indicating a better numerical value between the plot of the true positive rate against the false positive rate. Classification accuracy in the zero harmonic wave is 97.1%,

representing the ratio of correctly predicted instances to the total instances. The F1 score for KNN features of the same wave is 95.6%, reflecting a balance between precision (the

ratio of correctly predicted positive observations to the total predicted positives) and recall (the ratio of correctly predicted positive observations to all observations in the actual class).

On the contrary, the logistic regression of the ‘zero’ harmonic wave does not provide a good numerical value. This harmonic wave’s AUC is 46.9%, lower than the rate in the KNN feature. Similarly, the classification accuracy and F1 Score data values are lower than the ones for the KNN feature, at 55.1% and 61.3%, respectively.

Similarly, the neural network case for the harmonic ‘zero’ AC waves has an overall accuracy of 50 percent. However, metrics such as classification accuracy, F1 Score, precision, Matthew's correlation coefficient, and specificity fall slightly below the halfway mark at 47%, 49.5%, 36.3%, and 31.6%, respectively, indicating suboptimal accuracy values. On the other hand, metrics like area under the curve, recall, and logarithmic loss are marginally above 50% at 52.1%, 77.9%, and 61.8%, respectively. Despite these higher values, the neural network machine learning case for the harmonic ‘zero’ AC sinusoidal pattern is not significantly better than the KNN case.

Table 2: Harmonic “Zero” Predicted Score Curve

Model	AUC	CA	F1	Precision	Recall	MCC
kNN	1.000	0.996	0.994	0.988	1.000	0.994
Decision Tree	0.772	0.578	0.612	0.441	1.000	0.474
Logistic Regression	0.500	0.333	0.500	0.333	1.000	0.000
Neural Network	0.556	0.351	0.380	0.353	0.411	0.026
Constant	0.500	0.333	0.500	0.333	1.000	0.000

Table 2 displays the predicted values for these cases in the harmonic ‘zero’ AC (Alternating Current) sinusoidal waves. The KNN, Decision Tree, logistic regression, neural network, and constant machine learning cases have AUC values exactly or above 50%, at 100%, 77.2%, 50%, 55.6%, and 50%, respectively. While the

classification accuracy for KNN and decision trees is high at 99.6% and 57.8%, respectively, logistic regression, neural network, and constant cases have CA well below 50%, at 33.3%, 35.1%, and 33.3%, respectively. The Matthews correlation coefficient values for these cases, except KNN (99.4%), are significantly lower than the halfway accuracy values, indicating deficient performance in machine learning testing and

prediction of the harmonic AC sinusoidal patterns. In fact, the KNN model predicts 100% of the ‘zero’ harmonic AC waves accurately compared to the other cases. Overall, these cases for the predicted table show significant improvement than the test currently harmonic values as well.

Table 3: Harmonic “Zero” Error Data

	Logistic Regression	error	Neuron Network	error	KNN	error	Harmonic	Time	Voltage
1	0.33 → 0	0.667	0.38 → 0	0.615	0.33 → 0	0.952	0	0	0
2	0.33 → 0	0.667	0.38 → 0	0.616	1.00 → 0	0.994	0	5.00E-05	1.57073
3	0.33 → 0	0.667	0.38 → 0	0.617	1.00 → 0	0.284	0	0.0001	3.14108
4	0.33 → 0	0.667	0.38 → 0	0.617	1.00 → 0	0.779	0	0.00015	4.71065
5	0.33 → 0	0.667	0.38 → 0	0.618	1.00 → 0	0.200	0	0.0002	6.27905
6	0.33 → 0	0.667	0.38 → 0	0.618	1.00 → 0	0.582	0	0.00025	7.84591

Table 3 below displays the errors of the zero harmonic AC sinusoidal waves between the test score and the predicted score values in these machine learning

cases.

According to table 3, the logistic regression shows an increase in the probability cases by approximately 0.667 numbers for the first 6 sequence points. Similarly, the neural network error also increases by around 0.614. However, the KNN errors show slight differences across each of these points. Specifically, points 1, 2, and 4 exhibit a substantial increase in numerical value, while the rest of the points also show a slight increase.

These three datasets for ‘zero’ harmonic AC Sinusoidal wave patterns above are similar trend to one and two AC Sinusoidal waves. The changes of frequency and amplitude for these waves can accuracy vary their output even if they are sequentially continuous as their harmonic pattern.

4.3. MATLAB “One” Harmonics on Orange Data Mining Results

Table 4: Harmonic “One” Test Score Curve

Model	AUC	CA	F1	Precision	Recall	MCC	Spec	Log Loss
kNN	0.994	0.964	0.947	0.946	0.948	0.920	0.973	0.135
Tree	0.775	0.779	0.513	0.972	0.348	0.499	0.995	0.609
Logistic Regression	0.471	0.531	0.311	0.305	0.317	-0.0	0.638	0.637
Neural Network	0.418	0.634	0.072	0.231	0.042	-0.1	0.929	0.639
Constant	0.5	0.667	0.0	0.0	0.0	0.0	1.0	0.637
SVM	0.506	0.628	0.155	0.319	0.102	-0.010	0.891	0.637
Gradient Boosting	0.660	0.707	0.428	0.611	0.329	0.276	0.896	0.542

Table 4 illustrates the 'one' harmonic AC sinusoidal waves obtained by adding two odd numbers to their frequency and voltage amplitude. The machine learning KNN achieves 100% across various metrics, with the AUC, CA, F1 score,

precision, recall, MCC, and specificity reaching 99.4%, 96.4%, 94.7%, 94.6%, 94.8%, 92%, and 97.3%, respectively, mirroring the test score. However, the logarithm loss for this KNN ML case is slightly lower, like the 'zero' AC sinusoidal wave's case. In contrast, the CA, specificity, and logarithm loss for the neural network ML case in this harmonic AC sine wave exceed 50%, reaching 63.4%, 92.9%, and 63.9%, respectively, compared to other cases with lower values. The support vector machine ML case for the same harmonic wave shows a substantial increase in metrics such as AUC, CA, specificity, and logarithm loss, reaching 50.6%, 62.8%, 89.1%, and 63.7%, respectively. However, the remaining cases fall below the halfway range of 50%. Overall, compared to the 'zero' harmonic pattern, this harmonic AC sinusoidal wave exhibits significantly lower values, with only a slight decrease observed in the KNN case. The metrics for the other cases fluctuate between the two harmonic stages.

Table 5: Harmonic “One” Predicted Score Curve

Model	AUC	CA	F1	Precision	Recall	MCC	Spec	Log Loss
kNN	0.997	0.975	0.963	0.960	0.967	0.945	0.980	0.077
Tree	0.781	0.790	0.548	0.964	0.383	0.523	0.993	0.573
Logistic Regression	0.468	0.530	0.310	0.304	0.316	-0.0	0.637	0.637
Neural Network	0.511	0.643	0.358	0.446	0.299	0.128	0.814	0.619
Constant	0.5	0.400	0.470	0.333	0.799	-0.0	0.200	0.637
SVM	0.486	0.571	0.357	0.356	0.357	0.035	0.677	0.639
Gradient Boosting	0.725	0.749	0.444	0.849	0.301	0.400	0.973	0.511

The predicted values for the 'one' harmonic voltage AC sinusoidal waves are presented in table 5. Across all metrics of

AUC, CA, F1 score, precision, recall, and MCC, the KNN ML model shows a significant increase from the test score of the 'one' harmonic AC sine waves, with values of 100%, 99.6%, 99.7%, 100%, 99.4%, and 99.4%, respectively. Conversely, the values for the other ML model cases fluctuate in comparison to these results. Therefore, the predicted values for 'zero' and 'one' harmonic AC sinusoidal curves, particularly in the case of KNN, show substantial improvement from their test scores. Despite fluctuations in other cases, the predicted values demonstrate enhancement, indicating progress in controlling and optimizing of these harmonic curves.

4.4. MATLAB “Two” Harmonic on Orange Data Mining Results

"Two" harmonic AC sinusoidal waves exhibit an increase in both test scores and predicted scores compared to the values of "zero" harmonic AC sine curves. When the frequency and voltage amplitude vary across multiple AC sinusoidal curves, artificial intelligence and machine learning techniques can effectively control these harmonic waves under optimal conditions. Therefore, the harmonic "two" AC sinusoidal waves, depicted below, demonstrate an increment in values compared to both "zero" and "one" curves, respectively.

Table 6 displays the current test scores across various ML cases and models for the 'two' harmonic scenarios. As expected, the KNN ML model case maintains a score close to 100%. The values for the neutral network ML model and SVM align closely towards 50% with those of the 'zero' and 'one' cases, respectively.

Table 6: Harmonic “Two” Test Score Curve

Model	AUC	CA	F1	Precision	Recall	MCC	Spec	Log Loss
kNN	0.997	0.975	0.963	0.960	0.967	0.945	0.980	0.077
Tree	0.781	0.790	0.548	0.964	0.383	0.523	0.993	0.573
Logistic Regression	0.468	0.530	0.310	0.304	0.316	-0.0	0.637	0.637
Neural Network	0.511	0.643	0.358	0.446	0.299	0.128	0.814	0.619
Constant	0.5	0.400	0.470	0.333	0.799	-0.0	0.200	0.637
SVM	0.486	0.571	0.357	0.356	0.357	0.035	0.677	0.639
Gradient Boosting	0.725	0.749	0.444	0.849	0.301	0.400	0.973	0.511

Table 7: Harmonic “Two” Predicted Score Curve

Model	AUC	CA	F1	Precision	Recall	MCC
kNN	1.000	0.996	0.997	1.000	0.994	0.994
Tree	0.790	0.578	0.561	1.000	0.390	0.474
Logistic Regression	0.500	0.333	0.000	0.000	0.000	0.000
Neural Network	0.540	0.351	0.336	0.376	0.304	0.026
Constant	0.500	0.333	0.000	0.000	0.000	0.000

The predicted values for the 'two' harmonic AC sinusoidal curves

closely resemble those of the two previous harmonic scenarios. The KNN predicted values hover around 100%, while those for logistic regression and neutral network models are approximately 50%.

Table 8: Harmonic “One” Error Data

	Logistic Regression	error	Neuron Network	error	KNN	error	Harmonic	Time	Voltage
1	0.33 → 0	0.667	0.34 → 0	0.660	0.33 → 0	0.952	1	0	0
2	0.33 → 0	0.667	0.34 → 0	0.659	1.00 → 1	0.994	1	5.00E-05	1.64927
3	0.33 → 0	0.667	0.34 → 0	0.659	1.00 → 1	0.284	1	0.0001	3.29813
4	0.33 → 0	0.667	0.34 → 0	0.659	1.00 → 1	0.779	1	0.00015	4.94618
5	0.33 → 0	0.667	0.34 → 0	0.659	1.00 → 1	0.200	1	0.0002	6.593
6	0.33 → 0	0.667	0.34 → 0	0.659	1.00 → 1	0.582	1	0.00025	8.23821

The errors in the 'one' and 'two' harmonic AC sinusoidal values have increased substantially in the ML model's neutral network compared to the 'zero' harmonic data. However, the KNN and logistic regression values for these 'one' and 'two' harmonics are identical to each other, distinct from the 'zero' harmonic data, highlighting the importance of controlling these harmonics to mitigate power quality risks.

Table 9: Harmonic “Two” Error Data

	Logistic Regression	error	Neuron Network	error	KNN	error	Harmonic	Time	Voltage
1	0.33 → 0	0.667	0.28 → 0	0.725	0.33 → 0	0.952	2	0	0
2	0.33 → 0	0.667	0.28 → 0	0.725	1.00 → 2	0.994	2	5.00E-05	1.7278
3	0.33 → 0	0.667	0.28 → 0	0.724	1.00 → 2	0.284	2	0.0001	3.45518
4	0.33 → 0	0.667	0.28 → 0	0.724	1.00 → 2	0.779	2	0.00015	5.18171
5	0.33 → 0	0.667	0.28 → 0	0.723	1.00 → 2	0.200	2	0.0002	6.90696
6	0.33 → 0	0.667	0.28 → 0	0.723	1.00 → 2	0.582	2	0.00025	8.6305

4.5. Orange Data Mining Visualization on MATLAB Harmonic

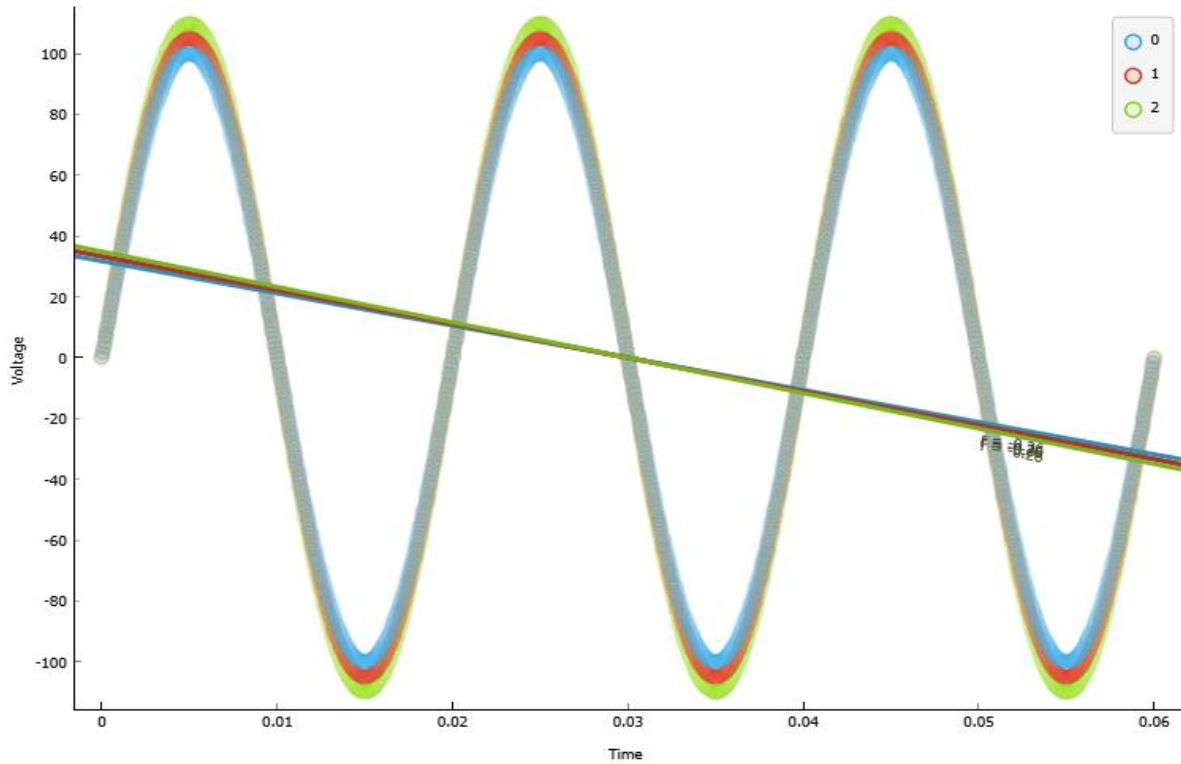


Figure 20: MATLAB Simulink Three Harmonics Diagram

Figure 20 illustrates the patterns of 'zero,' 'one,' and 'two' harmonic AC sinusoidal waves. The colors blue, red, and green represent the 'zero,' 'one,' and 'two' harmonic AC sinusoidal waves, respectively. The blue color represents the fundamental frequency (50 Hz) with a voltage amplitude of 100 Volts. The red and green colors depict additional frequencies and amplitudes added to the 'zero' harmonic wave, resulting in a composite pattern over the same time duration of 0.06 seconds. The regression line ($r = -0.26$) is a straight line intersecting these three harmonic curves, indicating their relationship, and illustrating the decreasing behavior in the voltage and time range, characteristic of electrical AC signals.

4.6. Regatron ACS Machine Harmonic on Orange Data Mining Results

The second way entails utilizing the Regatron ACS machine's harmonic data. When processed through the Orange Data Mining software, this data can be extremely disruptive because it is extracted using a hardware scenario rather than a software principle. As a result, the data and visualization performance may differ. Nonetheless, the hardware harmonic technique should be utilized with two parameters ('zero' - 100 V, 50 Hz, 'one' - 100 V, 5V, 50 Hz), rather from the software method, which has three parameters, as previously discussed. Figure 21 shows the specifications of the Regatron ACS Machine's computer vision. This software includes complex parameters, but I created with simple AC Sinusoidal harmonics and extracted their data.

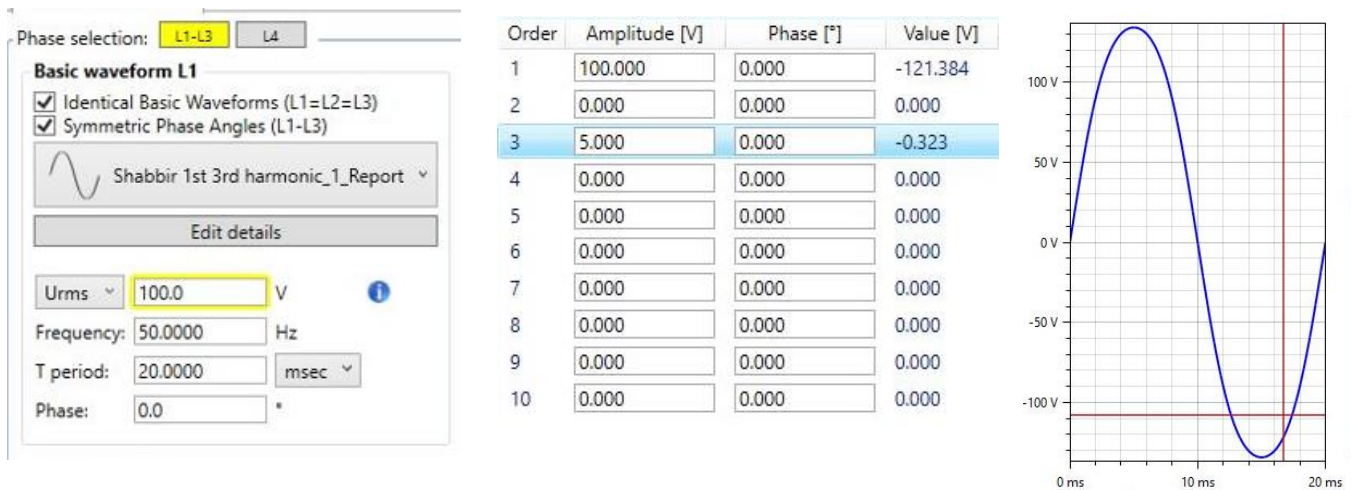


Figure 21: Regatron Machine's Software for Harmonics' Parameters

Table 10: Regatron Machine "One-Third" Harmonic's Test Score Results

Model	AUC	CA	F1	Precision	Recall	MCC	Spec	Log Loss
kNN	0.927	0.880	0.883	0.860	0.908	0.761	0.852	1.871
Tree	0.940	0.893	0.892	0.901	0.882	0.786	0.903	1.006
Logistic Regression	0.506	0.501	0.503	0.502	0.504	0.004	0.5	0.693
Neural Network	0.731	0.673	0.686	0.660	0.714	0.347	0.632	0.590
Constant	0.5	0.5	0.667	0.5	1.0	0.0	0.0	0.693
SVM	0.503	0.499	0.568	0.500	0.658	-0.001	0.340	0.693
Gradient Boosting	0.823	0.732	0.748	0.704	0.798	0.467	0.665	0.533

Table 11: Regatron Machine “One-Third” Harmonic’s Predicted Score Results

Model	AUC	CA	F1	Precision	Recall	MCC
kNN	0.962	0.921	0.924	0.894	0.956	0.845
Tree	0.984	0.930	0.929	0.941	0.918	0.861
Logistic Regression	0.506	0.502	0.503	0.502	0.503	0.004
Neural Network	0.737	0.674	0.685	0.663	0.708	0.349
Constant	0.500	0.500	0.667	0.500	1.000	0.000

According to Table 10, the Regatron ACS machine's harmonic data can be used for the machine learning model and the metric cases. When comparing both software and hardware values for the current test score, these values may be more accurate because of controlled percentage numbers if they are coming from the hardware method. The ML KNN model across all metric cases is higher than the 50% range. However, there might be a risk with the logarithm loss because the value exceeded 100% accuracy according to the Orange Data AI software. The values for the neutral network and SVM also show an increase compared to those from the software harmonic method, even though they fluctuate.

Table 12: Regatron Machine “One-Third” Harmonic’s Errors

	Logistic Regression	error	Neuron Network	error	KNN	error	Harmonic	Time	Voltage
1	0.50 → 0	0.500	0.57 → 0	0.433	1.00 → 0	0.000	0	0	0
2	0.50 → 0	0.500	0.57 → 0	0.428	0.67 → 0	0.333	0	5.00E-05	-2
3	0.50 → 0	0.500	0.58 → 0	0.424	0.33 → 1	0.667	0	0.0001	-4
4	0.50 → 0	0.449	0.60 → 0	0.403	1.00 → 0	0.000	0	0.00015	-20
5	0.50 → 0	0.500	0.59 → 0	0.414	0.33 → 1	0.667	0	0.0002	-8
6	0.50 → 0	0.449	0.60 → 0	0.404	0.33 → 1	0.667	0	0.00025	-24

The predicted values generated by the hardware harmonic method have shown improvement compared to their current test scores. Specifically, the metric cases, especially for the ML model's KNN, logistic regression, and support vector machine values, have demonstrated a significant enhancement in accuracy for the predicted AC

sinusoidal curves. This improvement suggests potential applications for optimizing future AC sinusoidal voltage waves within optimal ranges.

Table 12 shows the inaccuracies in the harmonic data of the Regatron ACS machine. The errors in logistic regression and neural network have the lowest numerical values, indicating a significant improvement from the current test result to the anticipated score. However, the error in KNN remains constant at 0.667 values between the current test result and the forecasted one

4.7. Orange Data Mining Visualization on Regatron ACS Harmonics

Figure 22 represents the AC sinusoidal curve for the Regatron ACS machine's harmonic data. Plotting this sinusoidal pattern has been more difficult due to the hardware condition. The root mean squared from the Regatron machine is $100 \times \sqrt{2}$ voltages due to the main grid voltages. The regression line shows a positive trend, unlike the software regression line. Hence, the voltage gap between the harmonics can expand instead of being in contact.

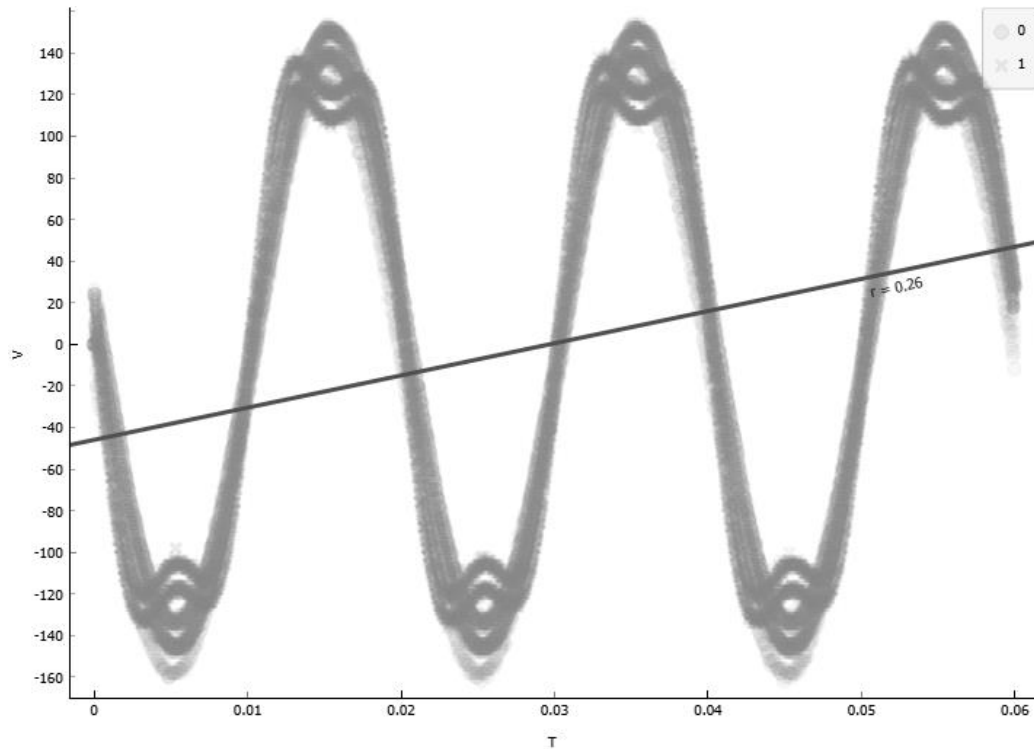


Figure 22: Regatron Machine's Two Harmonics Diagram

4.8. MATLAB “One-Set” Harmonics on Jupyter Notebook Results

The Orange Data Mining software significantly utilizes artificial intelligence and machine learning tactics to enhance accuracy and precision for the AC sinusoidal curve standard. However, the second software that offers even more precision for machine learning is Jupyter Notebook software. This software operates with Python Language, which is a bit easier compared to other computer programming languages. This type of computer programming language incorporates numerous blocks that correlate with machine learning algorithms. Some of these algorithms are therefore integrated into AC sinusoidal harmonic curves for detection, prediction, and forecasting of these waves ahead of the present period. If the detection of these harmonic waves is successful with the help of AI techniques, then numerous machines, power electronics, and electrical machineries can effectively control these harmonics, enhance power quality, and reduce performance costs.

MATLAB Simulink (one-set) can be attributed by testing only one result with three harmonic curves. More details of the one set MATLAB Simulink results are listed below.

For your information, Macro average refers to a method of computing the average performance metrics across multiple classes in a multi-class classification problem, while weighted average refers to a type of average where different values contributing to the average are given different weights based on their significance or importance. The analysis regarding the Jupyter Notebook’s predicted accuracies of these harmonics shows low performance. The support values are lower due to fewer harmonic points, making machine learning algorithms less effective. Except for the Decision Tree accuracy, which is almost one hundred percent accurate, all the other machine learning models have less than 50 percent accuracy. Consequently, there is low reliability in predicting harmonics in future AC Sinusoidal wavelengths, higher risks for electrical machineries and power electronics, which may function less cautiously.

MATLAB Simulink – 01 set Data

Accuracy: 0.43273231622746183 (Referred to Jupyter Notebook Software's results)

KNN Accuracy: 0.47

KNN Classification Report:

The predicted harmonic 'zero' and harmonic 'one' waves are below the 50% range. However, the metric case of recall and F1 score for the harmonic 'zero' wave is slightly above 50% compared to the 'one' harmonic wave.

Table 13: KNN Predicted Score

Harmonic	Precision	Recall	F1-Score	Support
0	0.47	0.65	0.55	260
1	0.38	0.44	0.41	225
Accuracy			0.47	721
Macro Average	0.53	0.47	0.46	721
Weighted Average	0.53	0.47	0.47	721

Decision Tree Accuracy: 0.99

Decision Tree Classification Report:

Table 14: Decision Tree Predicted Score

Harmonic	Precision	Recall	F1-Score	Support
0	1.00	0.99	1.00	260
1	0.97	0.99	0.98	225
Accuracy			0.99	721
Macro Average	0.99	0.99	0.99	721
Weighted Average	0.99	0.99	0.99	721

The decision tree for MATLAB Simulink shows values higher than those of the KNN ML results. Both 'zero' and 'one' harmonic waves are close to 100% in the metric factors of precision, recall, and F1 score values.

SVM Accuracy: 0.36

SVM Classification Report:

All the metric for the harmonic ‘zero’ and ‘one’ are below the 50% range for the predicted values

Table 15: Support Vector Machine Predicted Score

Harmonic	Precision	Recall	F1-Score	Support
0	0.37	0.49	0.42	260
1	0.27	0.16	0.20	225
Accuracy			0.36	721
Macro Average	0.34	0.35	0.34	721
Weighted Average	0.34	0.36	0.34	721

Logistic Regression Accuracy: 0.31

Logistic Regression Classification Report:

Table 16: Logistic Regression Predicted Score

Harmonic	Precision	Recall	F1-Score	Support
0	0.00	0.00	0.00	260
1	0.31	0.71	0.43	225
Accuracy			0.31	721
Macro Average	0.20	0.32	0.23	721
Weighted Average	0.19	0.31	0.22	721

The ‘one’ harmonic AC sine wave via recall metric are above the 50% predicted value while the rest of the values for both waves are below 50% ranges.

Neural Network Accuracy: 0.38

Neural Network Classification Report:

Table 17: Neural Network Predicted Score

Harmonic	Precision	Recall	F1-Score	Support
0	0.38	0.60	0.47	260
1	0.30	0.19	0.23	225
Accuracy			0.38	721
Macro Average	0.37	0.37	0.36	721
Weighted Average	0.37	0.38	0.36	721

The ‘zero’ harmonic curve are above the 50% range for the recall metric only and all of the values are below 50%. However, if the dataset for the harmonic can exceed into more data, then the accuracy will be precise and accuracy.

4.9. MATLAB “Ten-Set” Harmonics on Jupyter Notebook Results

MATLAB Simulink – 10 sets Data (Big Data)

Accuracy: 0.9954204829308909

KNN Accuracy: 1.00

KNN Classification Report:

Table 18: KNN Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	1.00	1.00	1.00	2390
1	1.00	1.00	1.00	2414
Accuracy			1.00	4804
Macro Average	1.00	1.00	1.00	4804
Weighted Average	1.00	1.00	1.00	4804

All of harmonic 0 and harmonic 1 are exactly 100% predicted values for all metric tasks.

Decision Tree Accuracy: 1.00

Decision Tree Classification Report:

Table 19: Decision Tree Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	1.00	0.99	1.00	2390
1	0.99	1.00	1.00	2414
Accuracy			1.00	4804
Macro Average	1.00	1.00	1.00	4804
Weighted Average	1.00	1.00	1.00	4804

Almost all of both harmonic values are 100% for these metric ranges

SVM Accuracy: 0.53

SVM Classification Report:

Table 20: SVM Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.52	0.71	0.60	2390
1	0.54	0.34	0.42	2414
Accuracy			0.53	4804
Macro Average	0.53	0.53	0.51	4804
Weighted Average	0.53	0.53	0.51	4804

Precision metric for harmonic 0 and harmonic 1 and the recall and F1 Score metrics for harmonic 0 are above 50% predicted values. The recall and F1 Score for harmonic 1 are below 50% values.

Logistic Regression Accuracy: 0.50

Logistic Regression Classification Report:

The harmonic 0 for all metric tasks and harmonic 1 wave's precision are slightly above 50%. Other values for harmonic 1 are below 50%.

Table 21: Logistic Regression Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.50	0.69	0.58	2390
1	0.50	0.31	0.38	2414
Accuracy			0.50	4804
Macro Average	0.50	0.50	0.48	4804
Weighted Average	0.50	0.50	0.48	4804

Neural Network Accuracy: 0.58

Neural Network Classification Report:

Table 22: Neural Network Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.55	0.94	0.69	2390
1	0.79	0.23	0.36	2414
Accuracy			0.58	4804
Macro Average	0.67	0.59	0.53	4804
Weighted Average	0.67	0.58	0.53	4804

Harmonic 1 wave's recall and F1 Score metric are below 50%. However, harmonic 0 curve's values for all metrics and harmonic 1 wave's precision are way above 50%.

The prediction sector study indicates that the accuracies of Random Forest (not shown in table – Thesis Case), KNN, and Decision Tree are nearly 100% accurate for the harmonic behavior scale, and therefore, they can partially control the harmonic. Furthermore, the values of the weighted and macro averages for Random Forest, Decision Tree, and KNN are the same. Gradient Boosting (not shown in table – Thesis Case) accuracy is 67 percent prediction, while Neural Network, SVM, and Logistic Regression accuracies are somewhat above 50 percent prediction. However, the macro and weighted averages of the gradient boosting and neural network accuracies are different. Because of the interference from the AC waves in MATLAB, the forecasting for these harmonic values is therefore less precise. In spite of these problems, the harmonics' forecast is more accurate

than it was in the beginning. This is explained by the fact that there is more data support now than there was in the first analysis. With additional data, these figures may, overall, become more accurate.

4.10. Regatron “Ten-Set” Harmonics on Jupyter Notebook Results

Collecting AC harmonic data from the equipment machine offers a higher probability of recreation, thus representing more realistic values compared to MATLAB simulations. However, the equipment machine introduces some errors into these harmonic waves, leading to undesired fluctuations. Additionally, external factors such as noise can influence these data points. Nevertheless, machine learning techniques applied to these data points are still capable of accurately determining their precision despite these challenges.

Regatron ACS Machine’s Harmonic – 10 set

Accuracy: 0.9125

KNN Accuracy: 0.87

KNN Classification Report:

All the values of the harmonic 0 and harmonic 1 across all the types of metrics are above 50% predicted range; an average values for improve the harmonic scale.

Table 23: Regatron KNN Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.88	0.86	0.87	2391
1	0.87	0.89	0.88	2409
Accuracy			0.87	4800
Macro Average	0.87	0.87	0.87	4800
Weighted Average	0.87	0.87	0.87	4800

Decision Tree Accuracy: 0.90

Decision Tree Classification Report:

Table 24: Regatron Decision Tree Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.90	0.90	0.90	2391
1	0.90	0.91	0.90	2409
Accuracy			0.90	4800
Macro Average	0.90	0.90	0.90	4800
Weighted Average	0.90	0.90	0.90	4800

Similar to KNN, all of these values for the harmonic 0 and harmonic 1 are above 50% predicted score; around 90% out of full score of predicted range.

SVM Accuracy: 0.64

SVM Classification Report:

Except for the harmonic 0 AC sinusoidal wave's recall metric, all of the metrics for both harmonic 0 and 1 are above 50% range of predicted score.

Table 25: Regatron SVM Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.71	0.47	0.57	2391
1	0.61	0.81	0.69	2409
Accuracy			0.64	4800
Macro Average	0.66	0.64	0.63	4800
Weighted Average	0.66	0.64	0.63	4800

Logistic Regression Accuracy: 0.50

Logistic Regression Classification Report:

Table 26: Regatron Logistic Regression Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.50	0.52	0.51	2391
1	0.51	0.49	0.50	2409
Accuracy			0.50	4800
Macro Average	0.50	0.50	0.50	4800
Weighted Average	0.50	0.50	0.50	4800

The table 25 shows the values of exactly 50% for the support vector machine; an average for predicted values.

Neural Network Accuracy: 0.67

Neural Network Classification Report:

The neutral network for the hardware implementation harmonic data are showing all values for harmonic 0 and harmonic 1. In fact, all of these values are above 50% predicted score – mediocre accuracy.

Table 27: Regatron Neural Network Predicted Score (10-Set)

Harmonic	Precision	Recall	F1-Score	Support
0	0.66	0.72	0.69	2391
1	0.69	0.63	0.66	2409
Accuracy			0.67	4800
Macro Average	0.68	0.67	0.67	4800
Weighted Average	0.68	0.67	0.67	4800

MATLAB Accuracy in comparison to smaller datasets, Simulink and the Regatron machine perform better with bigger harmonic data sets, resulting in more accurate detection. In most circumstances, greater datasets make machine learning methods more feasible and effective.

CHAPTER 5: SUMMARY OF RESEARCH WORK

The Orange Data Mining and Jupyter Notebook's Artificial Intelligence Software have demonstrated superior performance when analyzing electric harmonic pattern data. The term "harmonic" inherently introduces disturbances to power quality, causing it to deviate from the average standard for machinery. However, electrical machinery and power electronics can ultimately operate efficiently even after experiencing an initial drop in power quality. As engineers, it is imperative that we continually enhance these engineering machines and electronics.

Artificial intelligence and machine learning techniques provide a means to alleviate power quality risks. Moreover, optimizing controlled voltage units using AI software outputs for tested harmonics can enhance the efficacy of renewable energy sources. Furthermore, increased adoption of renewable energy facilitated by AI systems can lead to a reduction in carbon emissions from electrical engineering machines. Therefore, integrating this AI system into electricity management can yield benefits for both the future and electricity consumers.

Topics in Machine Learning and Deep Learning abound with algorithms applicable to real-life scenarios. Whether it is for weather forecasting, market analysis, or education, artificial intelligence techniques represent the future. These specific algorithms, based on electric harmonic AC sinusoidal waves, could serve as a solution for transitioning from conventional fuels to renewable energy sources, thereby reducing carbon emissions and enhancing power quality standards in the electric market.

CHAPTER 6: CONCLUSION

This research study focuses on the use of AI methodology to identify and analyze AC Sinusoidal Harmonic Waves. Given that all power quality waves contain harmonics, machine learning approaches proposition the most effective solution to the issues faced by electric equipment. Harmonics in Alternating Current systems can lead to multiple disturbances, such as voltage over-spikes and sags. These disturbances increase the cost of power for consumers also, impacting both reliability and affordability (SDG 07).

Third-world countries struggle to provide controlled electricity at a reasonable cost. Fossil fuels and petroleum-based generators (Crude oil from the underground and sea reservoirs) face harmonics issues when initially switched on. Similarly, hydroelectric dams when producing electricity from the flow of water to the turbines movement has experience disturbances in voltage and current due to harmonics. Despite the growing popularity of renewable energy, initial instabilities in running AC waves due to multiple non-user harmonics persist. These effects cause abnormal frequency and timing conditions in power electronics and transmission lines, potentially leading to faults and damage to electrical components.

As a result, managing power quality issues is crucial when harmonics are present. Artificial intelligence emerges as the most effective technique for controlling power disruptions while reducing costs. The future trend involves pairing AI and machine

learning algorithms with enhanced visualization, updated ML classification, and improved test data.

MATLAB Simulink software can detect harmonics by controlling frequency and voltage. Different frequency and voltage values in AC sinusoidal currents simulated in MATLAB yield harmonics data. This data can be analyzed using AI software such as Jupyter Notebook and Orange Data Mining, which employ machine learning algorithms to understand AC Sinusoidal Harmonic data. The KNN module is particularly effective in detecting points that simulate AC sinusoidal curves, while the NN module provides average values for harmonic patterns. SVM and logistic regression algorithms also test these harmonic waves, offering predictions for future use.

Although current machine learning percentages are low, predicting harmonics in future AC sinusoidal waves can help control this behavior and prevent faults, tripping electric lines, over-current transients, and difficulties in power electronics. This approach also manages three-phase electric motors and induction machines without excessive harmonics. Thus, using AI tools to improve power efficiency and manage harmonics in AC sinusoidal electricity at a reasonable cost is a viable strategy. Ultimately, AI and machine learning algorithms applied to AC sinusoidal harmonic waves present the best method for addressing these challenges.

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LIST OF PUBLICATIONS

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